

53-506 Introduction to English Linguistics

Martin Schweinberger

9. Februar 2015

WS 2014/15





Overview

Seminar Ia	2h Friday 14-16 (+ 2h Tutorial)
ESA K	AA-E1, ENG-1, AA1, LAA1, LAA14
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Overview

Course description

- ▶ This course will introduce students to the basic concepts and methods in the study and description of language. We shall be looking at the basic levels of linguistic analysis such as phonetics and phonology, morphology, syntax, semantics and pragmatics. We will also be looking into where these overlap and briefly address where linguistics cuts into other academic disciplines. The course will be taught in English.



Attendance

- ▶ You are allowed to miss two classes.
- ▶ If you miss only 2 classes, you do not need an excuse because you are officially allowed to miss 15% of a course!(cf. Modulhandbuch).
- ▶ If you miss more than 2 classes, you have to contact me!
- ▶ To pass this course you need to be present and participate regularly, and you have to pass the final exam.



Overview

Tutorials

- ▶ The tutorials help you understand the course content and will prepare you for the final exam. So: go to one of the tutorials!

Exercise/ work sheets ...

- ▶ will NOT be graded or corrected
- ▶ will also help you understand the course content and will prepare you for the final exam.

Literatur

- ▶ Berg (2013)
- ▶ Kortmann (2001)



Overview

Tipps

- ▶ Martin Hilpert (a linguist who also studied English Linguistics at Hamburg University) uploaded his videos showing his lecture "Intro to English linguistics" to YouTube (have a look: it is very good and I highly recommend it!)

<https://www.youtube.com/playlist?list=PLKgdsSsfw-fYCJ90tLikJRbXvI74g6FUw>



Overview

Levels of linguistic analysis

Semiotics	study of signs
Phonetics	study of sounds
Phonology	study of sound systems and interaction
Morphology	study of word structure
Syntax	study of sentence structure
Semantics	study of meaning
Pragmatics	study of meaning in context
Sociolinguistics/Language Change	study of variation in language
Psycho-linguistics	study of language and mind/brain



Overview

Session	Date	Topic
1	Fr, 17. Oct. 2014	Formalities and overview
2	Fr, 24. Oct. 2014	Language and the Brain
3	Fr, 31. Oct. 2014	History of the English Language
4	Fr, 7. Nov. 2014	Phonetics: Phones and Phonemes
5	Fr, 14. Nov. 2014	Phonology: Allophony and Syllables
6	Fr, 21. Nov. 2014	Morphology: Building Meaningful Blocks
7	Fr, 28. Nov. 2014	Syntax: Word Classes and Phrases
8	Fr, 5. Dec. 2014	Syntax: Sentence Structure and Semantic Roles
9	Fr, 12. Dec. 2014	Semantics: The Classical View
10	Fr, 19. Dec. 2014	Semantics: Prototypes and Categorization
11	Fr, 9. Jan. 2015	Pragmatics: Language in the Real World
12	Fr, 16. Jan. 2015	Sociolinguistics and Language Change: Different Voices
13	Fr, 23. Jan. 2015	Recapitulation
14	Fr, 30. Jan. 2015	In-class exam



Overview Microlinguistics



Phonetics and Phonology

Phonetics: study of sounds

Phonology: study of sound systems and sound interaction

- ▶ How do we produce sounds?
 - ▶ prəˌnʌnsiˈeɪʃn
<pronunciation>
 - ▶ lɔːˈnɔːdə
<law and order> (sounds like 'Laura Norder')
 - ▶ Dialect refers to differences on all levels of grammar, i.e. lexicon, morphology, syntax.
- ▶ What makes someone sound "German"?

Morphology

Morphology is the study of word structure

- ▶ What is a word?
- ▶ Sehirlilestiremediklerimizdensiniz.

(1) Sehir- li- les- tir- eme- dik- ler-
town- s/o.from become cause.to can't whom those
imiz- den- siniz
we one.of you.are

You are one of those whom we can't turn into a
town-dweller (Deutscher 2005: 24)

- ▶ Internal structure, e.g. disgraceful: dis#grace#ful
- ▶ Word formation: How do we "form"(new) words?



Syntax

Syntax is the study of sentence structure

- ▶ How do we combine words?

- (2) I like apples and bananas.

- (3) ?Apples I like and bananas.

- (4) *Like bananas apples I and.

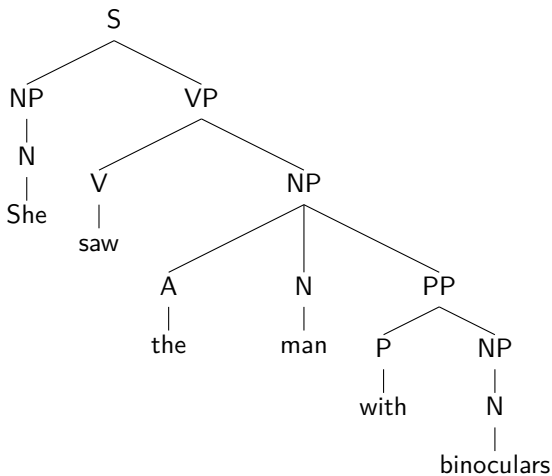
- ▶ Lexical and structural ambiguity:

- (5) She saw the man with the binoculars.

- (6) Time flies like an arrow. Fruit flies like a banana.

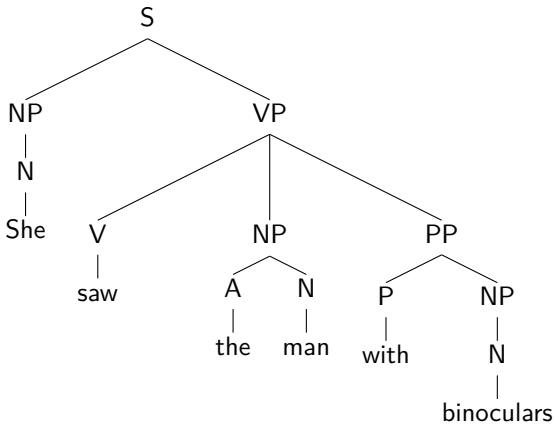


Syntax





Syntax





Semantics

Semantics is the study of meaning (relationships)

Features versus Prototypes

- ▶ Name 3 birds.
- ▶ Name 3 tools.
- ▶ Name 3 colors.



Semantics

Semantics is the study of meaning (relationships)

Features versus Prototypes

- ▶ Name 3 birds.
- ▶ Name 3 tools.
- ▶ Name 3 colors.

All of you had sparrow/pigeon, hammer/screwdriver, or red/blue.



Pragmatics (not microlinguistics)

Pragmatics is the study of meaning in context

What is meant by what is said?

- ▶ Can you take out the trash.
- ▶ It's warm in here.
- ▶ I wouldn't do that.

In all three cases, the literal meaning deviates from the intended message.



What is Language?



Discuss in groups of three:

- ▶ What is Language/a language?
- ▶ Language vs. dialect vs. accent?
- ▶ Provide a brief definition for each.
- ▶ How many languages are there?



What is Language?

Discuss in groups of three:

- ▶ What is Language/
a language?
 - ▶ Continuum, hard to separate!
- ▶ Language vs. dialect vs. accent?
 - ▶ Accent refers to differences in pronunciation, only.
- ▶ Provide a brief definition for each.
 - ▶ Dialect refers to differences on all levels of grammar, i.e. lexicon, morphology, syntax.
- ▶ How many languages are there?
 - ▶ App. 6,000 (depending on what counts as a "language")





What is Language?

Language is a system of visual, auditory, or tactile symbols (arbitrary signs) used in human communication and the rules used to manipulate them.

- ▶ Human language differs from animal communication.
- ▶ Only humans use language.
- ▶ All human cultures use language.
- ▶ Every child learns language.
- ▶ Language is handled by certain parts of the brain.
- ▶ All languages share certain features.
- ▶ All languages have certain regularities (grammar: word order).

...



What is Language?

Language is a system of visual, auditory, or tactile symbols (arbitrary signs) used in human communication and the rules used to manipulate them.

- ▶ Language is social.
- ▶ Language always changes.
- ▶ All languages are related.
- ▶ Language is creative.

...



Language and Communication

Human language vs. animal communication

- ▶ But don't animals communicate too?
- ▶ What is so different about human communication?
- ▶ How do we differentiate between a dog's bark, a bee dance and human communication?
- ▶ Kanzi!





Features of Language

- ▶ Displacement
 - ▶ Momentary and immediate meaning $\neq$ ability to refer to past and/or future. (bee communication?)
- ▶ Arbitrariness
 - ▶ Usually no natural connection between a linguistic form and its meaning. Relation is arbitrary (willkürlich), not iconic. (onomatopoeia? kikiriki)
- ▶ Productivity
 - ▶ Humans are frequently creating new expressions to refer to novel experiences, objects, or events. (bees: cannot communicate vertical distance).



Features of Language

- ▶ Traditional transmission
 - ▶ Genetic acquisition and traditional transmission.
 - ▶ Korean infant will speak English, if adopted by L1 English speakers.
 - ▶ Some animals are born with a distinct set of signals that they produce instinctively (cat: meow).
- ▶ Duality
 - ▶ Language is organized at two levels simultaneously.
 - ▶ Level of physical production of distinct sounds and the level of meaning (interpreting combination of sounds).



Linguistics and Semiotics

Linguistics

- ▶ Linguistics is the scientific study of systems of symbols, i.e. of language or particular languages, and of their rules (grammar) used by humans to communicate and express ideas and feelings.

Semiotics

- ▶ Semiotics is the scientific study of signs, sign systems of and sign processes in general, i.e. not restricted to verbal, human signs.

Semiotics and Signs



Signs

- ▶ A sign is sth. that stands for sth. else:
aliquid stat pro aliquo
- ▶ Form
 - ▶ Sound pattern, gesture, picture, writing, fashion, hair cut, ...
- ▶ Function
 - ▶ Meaning, concept, mental representation



I will be here tomorrow.





Types of Signs

Indexes ("pointing towards")

- ▶ Here, there, I, him, that. . . etc.
(note: there is still a symbolic/conventional element: ich, moi, lui etc.)

Icons (similarity)

- ▶ Onomatopoeia ("Lautmalerei")
E.g. *splash, crash, swoosh; that movie was so booooooring.*
- ▶ These are very rare examples and context- and culture-specific.

Symbols (arbitrariness)

- ▶ Most of language is symbolic; i.e. conventionalized and has to be learned.



Saussure's sign

- ▶ Ferdinand de Saussure (1857-1913; Swiss linguist)
 - ▶ Contemporary linguistics is, to a large extent, still based on concepts developed by him.
 - ▶ His main ideas were published posthumously from his lecture materials (1916) under the title "Cours de linguistique générale" (Course in General Linguistics).
 - ▶ He can be considered the founder of modern linguistics.
 - ▶ His school of thought is referred to as "Structuralism" (he is still an important figure in structuralist thought)



Saussure's sign

- ▶ Ferdinand de Saussure
(1857-1913; Swiss linguist)
 - ▶ Introduced the distinction between ...
 - ▶ Synchronic linguistics
(study of language at a certain time)
 - ▶ Diachronic linguistics
(study of language change over time)
 - ▶ Introduced the "primacy of the spoken word"
 - ▶ Shift away from a purely diachronic perspective to a focus on synchronic language use.
 - ▶ Fostered the transition from the prescriptive period to descriptive approach towards language.



Saussure's sign

- ▶ Ferdinand de Saussure
(1857-1913; Swiss linguist)
 - ▶ Focus on the structure of language.
 - ▶ Language system (langue) is the center of study, not the concrete language use (parole).
 - ▶ Principle of Arbitrariness

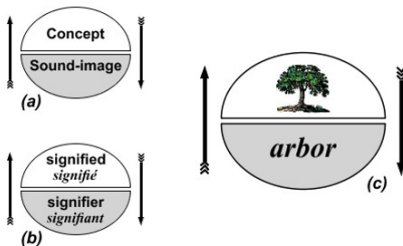


There is no internal link between a sound pattern and the meaning of a concept!

- ▶ The concept of a dog is represented by different sound patterns depending on the linguistic code used to refer to a dog and, thus, depends upon social convention.

Saussure's sign

- ▶ Ferdinand de Saussure (1857-1913; Swiss linguist)
 - ▶ Each linguistic sign (word) consists out of a sound sequence (signifier) and a concept (signified)
 - ▶ The relationship between the sound sequence and the concept of a linguistic sign is arbitrary and depends upon social convention.



Bühler's "Organon Model"

- ▶ Karl Bühler (1879-1963)
 - ▶ approached language by examining the (possible) functions of language.
 - ▶ Father of "Functionalism"
 - ▶ The "Organon Model" is a functional description of sign usage/communication.
 - ▶ Function(s) of language according to Bühler
 - ▶ expressive function (express feelings/beliefs)
 - ▶ representative function (talk about the world)
 - ▶ appellative function (e.g. make a request).

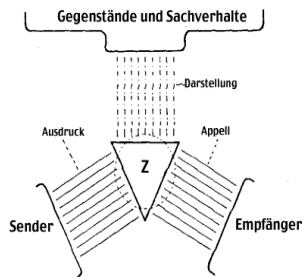


Fig. 3.





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Language and the Brain

History of English

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Semantics

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Sociolinguistics

Recapitulation



Introduction to English Linguistics

Language and the Brain

Today's topics and terms

- ▶ Aphasia
- ▶ Language area
- ▶ Critical period and plasticity



Language and the Brain

Localizationism

- ▶ Brain consist of compartmentalized and specialized areas called "modules"
- ▶ Modules are specialized neurons in certain regions dealing with simpler tasks (e.g. exclusively language-related tasks)

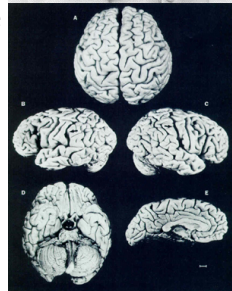
Connectionism (holism)

- ▶ Language related abilities are carried out by larger parts of the brain
- ▶ Focus on how language is dependent on other, broader cognitive abilities (memory, abstract thinking, attention, ...)
- ▶ No language centers but "network nodes"



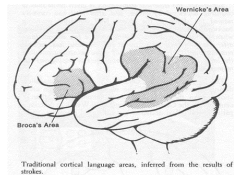
Localizationism

- ▶ Paul Broca (1824-1880)
- ▶ Brain has two hemispheres
- ▶ "Damage to the left hemisphere is more strongly correlated with language impairment than damage to the right hemisphere."
- ▶ Is there a language module in the left hemisphere?



Language impairments (aphasia)

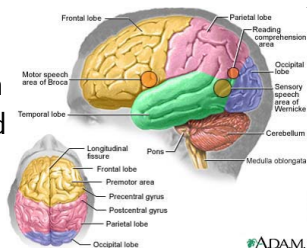
- ▶ Language deficits acquired after brain injury.
- ▶ Some aphasiacs perform normal on non-verbal tests of IQ, they have neither trouble cooking, nor walking a complex route home.
- ▶ Aphasias are caused by damage to certain regions of the left hemisphere (language area) Researchers discovered different kinds of aphasias
- ▶ Different symptoms correspond to different regions being impaired:
 - ▶ more to the front: speech production.
 - ▶ more to the back: comprehension.





Broca's aphasia

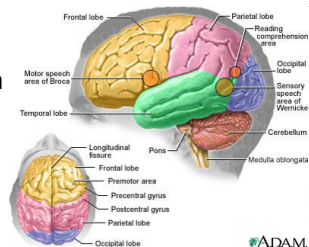
- ▶ In 1861, Broca presented data from a patient called Tan Tan. Tan could say this one syllable ("tan") and repeated it if asked to, but seemed surprised that he couldn't get his message across. Tan's comprehension was in fact good.
- ▶ Postmortem examination showed a brain lesion at the lower areas of the frontal lobe (Broca's region)
- ▶ Symptoms
 - ▶ fluent, slow, deliberate, effortful, omission of grammatical markers (e.g. "Boy go store" instead of "The boy has gone to the store")





Wernicke's aphasia

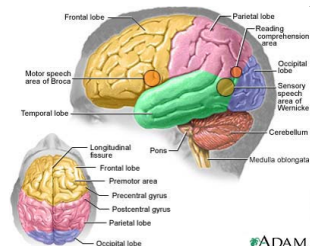
- ▶ 1874, Carl Wernicke presented data of two patients whose symptoms were quite unlike Tan's.
- ▶ Speech was fluent (good intonation), but contained unusual semantic features (circumlocutions, phonemic substitutions, neologisms).
- ▶ Comprehension was severely impaired.
- ▶ Lesion was situated at the back and top of the temporal lobe (Wernicke area).
- ▶ Symptoms
 - ▶ Very fluent, normal use of grammatical markers, mis-selection of words ("wine" for "why"), lack of meaningful content.





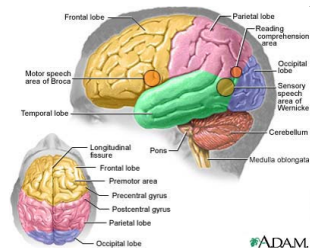
Anomic aphasia

- ▶ All aphasiacs have anomia of some kind, i.e. problems remembering the names of things.
- ▶ In 1993, cognitive psychologist Ashcraft experienced a temporal anomia.
 - ▶ temporarily unable to remember the names of things he frequently used or referred to.
- ▶ Caused by small lesions (little strokes) within the language area (anterior left temporal lobe).
 - ▶ Syntax remains unimpaired and comprehension is quite spared, difficulty finding specific substantive words.



Other aphasias

- ▶ Conduction aphasia
 - ▶ Inability to repeat spoken language, but only little comprehension and/or production difficulty.
- ▶ Pure word deafness
 - ▶ Inability to make sense of language.
- ▶ Transcortical motor aphasia
 - ▶ Patients will use fragmentary or little language.
 - ▶ Repetition; grammar and comprehension are spared.
- ▶ Transcortical sensory aphasia
 - ▶ Poor comprehension, but fluent, semantically empty speech.





Language impairments (aphasia)

- ▶ Different aphasias are linked to damage in different brain areas of the central left hemisphere
 - ▶ Problems in coming up with lexical items (words): mild damages in the language area around the Sylvian fissure.
 - ▶ Problems producing language sounds correctly and generating syntactic strings are connected with lesions in anterior brain regions (Broca's area).
 - ▶ Problems with comprehension and "empty speech" are associated with Wernicke's area.
 - ▶ Problems with repetition result from damage to area between Broca's and Wernicke's area.



We have gained knowledge about the language areas from
aphasias.

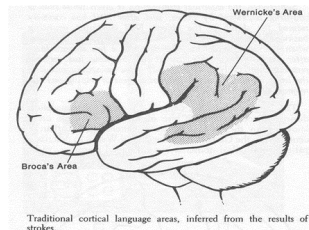
BUT

Which other ways of finding out where certain abilities are
performed do we have?



Detecting brain activity

- ▶ Cortical Stimulation
 - ▶ Putting electrodes into brains and stimulating certain regions, then asking the subjects what they experienced or describing what behavior they exhibited.
- ▶ PET scans (Positron Emission Tomography)
 - ▶ Shows brain activation during certain tasks
- ▶ CAT-scans (Computerized Axial Tomography)
- ▶ MRIs (Magnetic resonance Imaging)

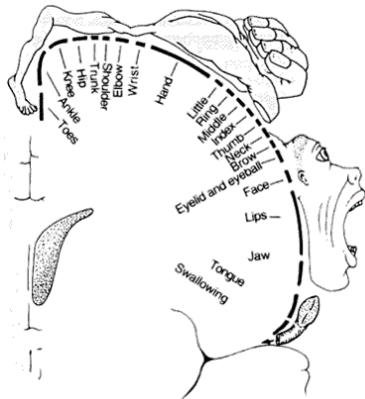
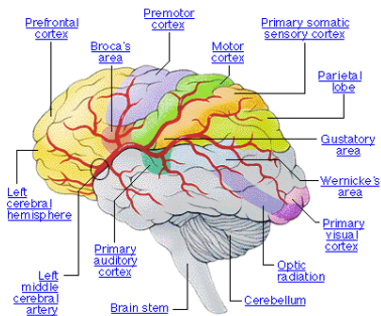




Nerve cells both in the cortical structure [surface of the brain] and the subcortical [inner] areas of the hemispheres of the brain are involved in both producing and understanding language. Within the left hemisphere, a "language area" can be delimited that includes areas right next to the primary motor areas of the brain [parts responsible for triggering muscular movement] and the primary sensory areas [sensing and feeling], as well as areas further back involved in taking in information presented either visually or auditorially. (Obler and Gjerlow 1999: 26)



Regions of the brain





Which other kinds of evidence indicate that the left hemisphere is dominant?



Left-hemisphere dominance for language

- ▶ Wada test
 - ▶ Anesthetizing one brain-hemisphere so subjects cannot speak for several minutes (sound like aphasics a few minutes after that)
- ▶ Tachistoscopic presentation
 - ▶ Visual stimuli are presented to one eye (hemisphere/-field) only (error rate and speed of response)
- ▶ Dichotic listening technique
 - ▶ Different auditory stimuli presented to both ears (error rate of dominant hemisphere is lower)



Left-hemisphere dominance for language

Split brain patients

- ▶ Connection between hemispheres is severed. Patients cannot name objects they don't see but feel with their left hand.
- ▶ BUT patients can name them if they don't see them but feel them with their right hand (left hemisphere!)





Critical period and plasticity

The critical period

- ▶ Children are not born with fully developed linguistic abilities.
- ▶ Lateralisation
 - ▶ One-sidedness process begins in early childhood and lasts until puberty
- ▶ During lateralisation the human brain is most ready to receive input and learn a particular language. This phase is called critical period.
- ▶ Birds have it as well!
- ▶ Neurons grow dendrites and synapses. If dendrites are not activated, they "dry up".



Critical period and plasticity

Genie

- ▶ 1970, a girl of 13 called Genie was admitted to a hospital in Los Angeles.
- ▶ Genie spent most of her life in a small closed room, basically without linguistic input (only mother came to her every day for a few minutes to deliver food). The linguistic ability of Genie was immensely underdeveloped – basically she couldn't use language. After a short while she began to imitate sounds and managed to acquire limited linguistic abilities (speaking and understanding a fair amount of English words).



Critical period and plasticity

Genie

- ▶ Genie never managed to form grammatically complex sentences!
- ▶ She had no "left sphere language facility"!
- ▶ Genie was using the right hemisphere for language functions (dichotic listening test).
- ▶ During speech acquisition Genie went through the same phases "normal" children go through during speech and language acquisition.



Critical period and plasticity

Mental lexicon

- ▶ We can infer from certain phenomena how our mental lexicon is organized (e.g. tip of the tongue phenomenon, slip of the tongue).
- ▶ Mental lexicon
 - ▶ The mental lexicon contains all our concepts and the words they are denoted.
 - ▶ It also provides information about how these words are related to others, e.g. antonymic relation (opposition): hot - cold; hyperonymy and hyponymy (superordinate/subordinate relation): fruit - orange; rhyme (similar phonological properties): lime, mine, fine

...



Critical period and plasticity

Mental lexicon

- ▶ Tip of the tongue phenomenon
 - ▶ The feeling that we know a word but just cannot "find" it momentarily. Sometimes we know how the word sounds, how many syllables it has, with which letter it begins, or with which words it rhymes.
 - ▶ Speakers produced *sectant*, *sextet*, *sexton* when asked to name a particular type of navigational instrument (*sextant*).
 - ▶ Others said *fire distinguisher* instead of *fire extinguisher* or *transcendental medication* instead of *transcendental meditation*.
 - ▶ malapropisms: "near misses"



Critical period and plasticity

Mental lexicon

- ▶ Slip of the tongue phenomenon (Spoonerisms)
 - ▶ William Spooner (Oxford Anglican clergyman)
 - ▶ "You have hissed all my mystery lectures".
 - ▶ (Initial) sounds or words are interchanged:
 - ▶ "Long shory stort", "use the door to open the key",
"fifty-pound dog of bag food", "beel fetter", "stick neff"
 - ...
- ▶ Commonly treated as errors of articulation but they are rather slips of the brain as it tries to organize linguistic messages.



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Today's topics and terms

- ▶ Family tree and Comparative Reconstruction
- ▶ Diachronic linguistics
 - ▶ Old English
 - ▶ Middle English
 - ▶ Early Modern English
 - ▶ Modern English
 - ▶ (Present day English)



Orthography and Pronunciation

ghoti

- ▶ "fish"
- ▶ enough, women, nation

seagh

- ▶ "chef"
- ▶ sure, dead, laugh

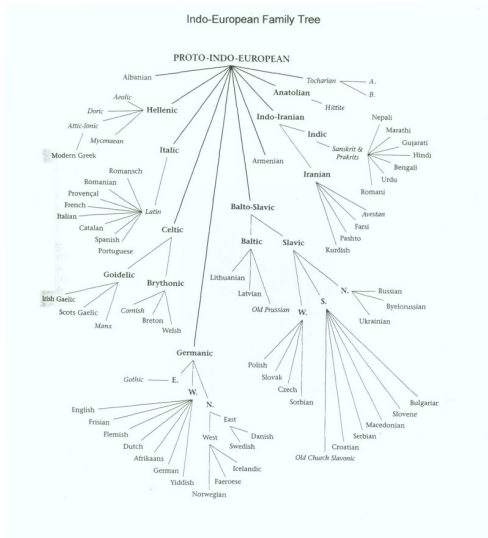
ghoughphtghtteeau

- ▶ hiccough, dough, phthisis, neighbour, gazette, plateau
- ▶ "potato"



Family tree

- ▶ Similar features indicates that both Latin and Greek are descendants of Sanskrit.
 - ▶ The Sanskrit language, whatever be its antiquity, is of a wonderful structure, more perfect than the Greek, more copious than the Latin, and more exquisitely refined than either, yet bearing to both of them a stronger affinity, both in roots of verbs and in the forms of grammar, than could possibly have been produced by accident. (Sir William Jones a British government official in India 1796; quoted from Yule 2006: 182)
- ▶ During 19th century linguists inferred that there existed a language that was the common ancestor of most languages in India and Europe about 4000 BC:
(Proto-) Indo-European.





Family connections

- ▶ Relation between Italian and Hindi?
- ▶ Taking a look at earlier forms of languages (Latin and Sanskrit)

Sanskrit	Latin	Ancient Greek	Translation
pitar	pater	pater	father
bhratar	frater	phrather	brother



Cognates

- ▶ Similar form of words in different languages that have a very similar meaning.

	<mother>		<Mutter>	
English	<father>	are cognates of	<Vater>	German
	<friend>		<Freund>	

- ▶ Inference: Modern English and Modern German probably have a common ancestor.



Comparative reconstruction

- ▶ Procedure employing cognates to reconstruct the original or proto-form in the common ancestral language:
 - ▶ Majority principle
 - ▶ If, in a cognate set, three words begin with a "p" and one with a "b", probably the majority has retained the original initial whereas the minority has changed.
 - ▶ Most natural development principle
 - ▶ Some sound changes are very common, others quite unlikely.
- 1. Final vowel often disappears: vino → vin
- 2. Voiceless sounds become voiced (typically between vowels): muta → muda
- 3. stops become fricatives: ripa → riva
- 4. Consonants become voiceless at the end of words:
rizu → ris



Sound reconstruction

Languages			
A	B	C	Translation
cantare	cantar	chanter	("sing")
catena	cadena	chaine	("chain")
caro	caro	cher	("dear")
cavallo	caballo	cheval	("horse")

- ▶ A and B begin with a [k] while C is the only one beginning with [ʃ].
- ▶ Initial sound [k] in A and B is probably older than [ʃ] in C.
- ▶ Most natural development principle supports the hypothesis. that Latin is the common ancestor:
cantare, catena, carus, caballus → [k]



Word reconstruction

Languages				
A	B	C	Protoform	Translation
mube	mupe	mup	?	("stream")
abadi	apati	apat	?	("rock")
agana	akana	akan	?	("knife")
enugu	enuku	enuk	?	("diamond")

- What are the protoforms/which language is the most archaic?



Diachronic linguistics

Language change over time

- ▶ Earlier forms of English differ substantially from modern English
- ▶ Internal factors for language change
 - ▶ Vowel shortening and lengthening because trochaic feet patterns were being selected for
- ▶ External factors for language change
 - ▶ Influx of new lexemes due to language contact (Viking settlements, Norman Conquest)



The Lord's Prayer through time

- ▶ Fæder ure þu þe eart on heofonum; Si þin nama gehalgod to becume þin rice gewurþe ðin willa on eorðan swa swa on heofonum.
(Old English, c.1100)
- ▶ Oure fadir that art in heuenes, halewid be thi name; thi kyndoom come to; be thi wille don in erthe as in heuene.
(Middle English, c.1380)
- ▶ Our Father which art in heauen, hallowed be thy Name. Thy kingdome come. Thy will be done euen in earth, as it is in heauen.
(Early Modern English, 1602)
- ▶ Our Father, who art in heaven, Hallowed be thy Name. Thy kingdom come. Thy will be done, On earth as it is in heaven.
(Modern English, 1928)
- ▶ Our Father in Heaven, let your holy name be known, let your kingdom come, and your will be done, on earth as in heaven.
(Modern/Contemporary English, 1970)



Pre-English (before 450)

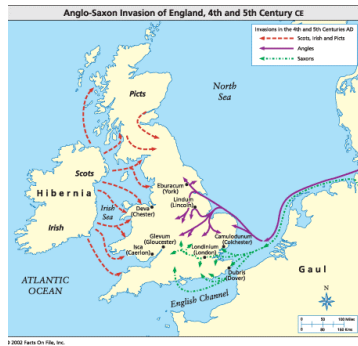
- ▶ At first, England was inhabited by the Celts
- ▶ Celtic language was spoken.
- ▶ Some traces of Celtic origin is preserved in certain place names
- ▶ 43-410 Romans came to England and Latin became the official language
- ▶ Latin (of that period) had a very weak influence on the development of the English language



Old English (450-1150)

- ▶ Germanic tribes arrive in England from the middle of the 5th century
 - ▶ Angles
 - ▶ Saxons
 - ▶ Jutes and Frisians
- ▶ Germanic word-stock is preserved in basic terms

(7) man: mann, woman: wif, child: cild, house: hus, food: mete, eat: etan, drink: drincan, fight: feohtan, etc.)





Old English (450-1150)

- ▶ Germanic was fully inflected (like modern German)
- ▶ Four major dialects
 - ▶ West Saxon (most texts that were preserved are written in West Saxon)
 - ▶ Kentish
 - ▶ Anglian
 - ▶ Mercian (plus Northumbrian)





Old English (450-1150)

- ▶ Germanic tribes were pagans as we can still detect in English weekdays (Woden and Thor).
- ▶ From the sixth to the eighth century the Anglo-Saxons were converted to Christianity.
- ▶ Latin (language of religion) came to influence English
 - (8) angel, bishop, candle, church, martyr, priest, school, ...

Old English and Old Norse

- ▶ From the eighth through the 9th and 10th century a northern European tribe, the Vikings, came to England, at first to plunder, then to settle.
- ▶ Old Norse, the language of the Vikings, still has some influence on Present-day Modern English.
- ▶ Mostly the influence of Old Norse can be seen in place and family names.
- ▶ Forms that originate from Old Norse and were preserved:

(9) give, law, leg, skin, sky, take, they





Middle English (1150-1500)

Norman Conquest in 1066

- ▶ William the Conqueror victory in Hastings
- ▶ French became the language of the ruling class and the language of the nobility, the government, the law, and the civilized life in England.
- ▶ Enormous influx of (Norman) French vocabulary

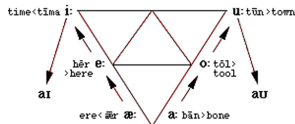




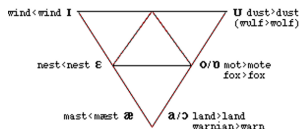
Middle English (1150-1500)

The Great English Vowel Shift

- ▶ Leveling of inflection increases
- ▶ Great English Vowel Shift
- ▶ Major change of the English vowel system started app. 1450 and lasted until app. 1750.
- ▶ Affected all long vowels of Middle English (ME).



Retention of Short Vowels

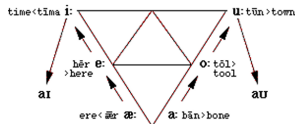




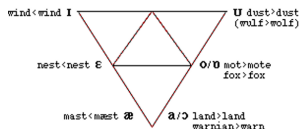
Middle English (1150-1500)

The Great English Vowel Shift

- ▶ Long vowels
 - ▶ raised or diphthongized
 - ▶ long [o:] in ME fode (food) was raised to [u:],
 - ▶ long [i:] in ME child and lyf (life) was diphthongized to [ai].
- ▶ Other effects
 - ▶ Short high back rounded vowel [ʊ] was rounded and centralised to [ʌ]; but [bʊt] became [bʌt].



Retention of Short Vowels





The Great English Vowel Shift

Vowel	Chaucer	Translation	Shakespeare
/i:/	[fi:f]	five	[faɪf]
/e:/	[me:də]	meat	[mi:t]
/ɛ:/	[kle:nə]	clean	[kle:n]
/a:/	[na:mə]	name	[ne:m]
/ɔ:/	[gɔ:tə]	goat	[gɔ:t]
/o:/	[ɹɔ:tə]	root	[ru:t]
/u:/	[du:n]	down	[daun]



Early Modern English (1500-1700)

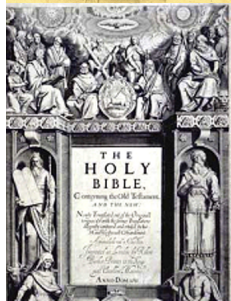
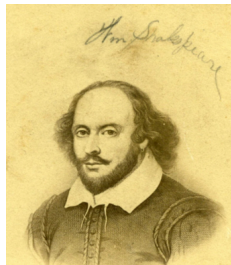
British expansion

- ▶ Introduction of printing to England in 1476 by William Caxton
- ▶ Standardization and Regulation of spelling
- ▶ Main phase of the GVS
(fundamental change of the vowel system)
- ▶ The spread of English around the world starts with the discovery of the New World and the race for colonies and resources.

Early Modern English (1500-1700)

British expansion

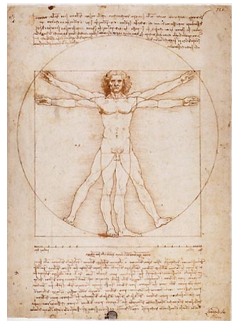
- ▶ Increase of the vocabulary
- ▶ Large-scale borrowing from Latin, Greek, French, and other European tongues
- ▶ First English-language dictionaries
- ▶ Most important literary influences on Early Modern English:
 - ▶ William Shakespeare (1564-1616)
 - ▶ King James Bible (1611)



Early Modern English (1500-1700)

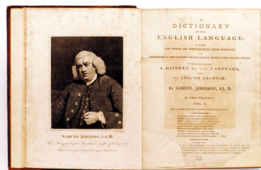
Renaissance

- ▶ Renewed interest in classical texts and languages.
- ▶ Enrichment of language and ideas through explorations, colonization, findings, and inventions in early forms of empirical science.
- ▶ While some viewed the enrichment of English by foreign vocabulary fruitful, others objected against this borrowing and spoke for a "pure" and "unmixed" English without "inkorn terms".



Modern English (1700 - ...)

- ▶ By 1700 English has (nearly) reached its present form.
- ▶ Increasing standardization
- ▶ Samuel Johnson's Dictionary of the English Language.
- ▶ Present Day English is considered to be one kind of Modern English.
- ▶ Analytic language (Modern English = period of lost inflection)
- ▶ Relation of words is indicated by a relatively fixed word order.





Modern English (1700 - ...)

- ▶ ModE's lexicon can be assigned to three different origins:
 - ▶ Germanic words that survived since OE
 - ▶ Vocabulary adopted from Latin, French and other European tongues
 - ▶ Words that have their origin in the geographical expansion of the British Empire
- ▶ Pronunciation is much the same as in EmodE
- ▶ Rise of Received Pronunciation in the early 20th century
- ▶ Received Pronunciation is an accent that, concerning linguistics, serves as reference in phonetics and phonology.
- ▶ At the beginning of the 21st century, the regional varieties exhibit an increase in independence and begin more and more to form their own "standards".



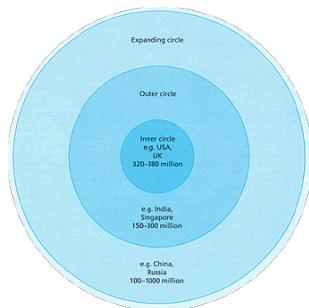
English as a Global Language

- ▶ Up to 1200 English was limited to the British Isles
- ▶ Phases of expansion
 - ▶ Spreading of English across the British Isles
 - ▶ Spreading beyond the British Isles due to colonization!
- ▶ Native language of about 340.000.000 people
- ▶ 2nd language in about 60 countries
- ▶ Major global language



English as a Global Language

- ▶ English today is the most widely used language
- ▶ Braj Kachru accounts for the different functions English has in different territories by introducing the model of the three circles of English.
 - ▶ 1st circle: English is native language
 - ▶ 2nd circle: English is second language
 - ▶ 3rd circle: English is a foreign language, but is widely understood





English as a Global Language

- ▶ English is the first truly international language
- ▶ Its the global lingua franca (also in intra-national communication, e.g. India)
- ▶ Internationally and cross-culturally there are various domains where English is used as lingua franca:
 - ▶ Economy
 - ▶ Politics
 - ▶ Science
 - ▶ International travel
 - ▶ Trade and transport
 - ▶ Modern technology (computer and Internet related terminology)



The Future of English

- ▶ Former third world countries in South Asia and Africa, especially grow rapidly by economic standards and in terms of population
- ▶ The number of speakers in highly developed countries (Western culture) is by comparison in decline
- ▶ In the future we will probably see an increase in independence of varieties which are still regarded as "inferior" sub-varieties of English



The Future of English

- ▶ Will English be substituted by another language, e.g. Mandarin (Chinese)?
 - ▶ Well, we don't know, but . . .
 - ▶ English is the established lingua franca and it appears to be rather unlikely that this will change anytime soon.
 - ▶ Countries where English is L2 are among those nations that gain importance (India, Philippines, Tanzania, Nigeria, . . .).
 - ▶ Pidgin and creole varieties will become more widespread and will gain prestige (accepted alternative standards)



The Future of English

- ▶ Do we need a universal language?
 - ▶ Between 1880 and 1907 53 universal languages were proposed (thousands throughout our attested history)
 - ▶ Most universal languages are forgotten (lack of speakers)
 - ▶ Recent history has shown language policy to be a highly emotional issue, the language of a country often symbolizing its independence and nationalism. (Baugh and Cable 1991: 7)



Orthography and Pronunciation

Different spellings for same sound, e.g. for [i:]

<ae> Caesar	<eo> people	<ey> key
<ay> quay	<ea> sea, tea	<i> ski, police,
<e> be	<ee> sneeze	<ie> field, yield
<ei> seize	<oe> Phoenix	

Same spelling for different sounds, e.g. <ea>

[eə] bear	[ɑ:] heart
[Iə] beard	[e] head, dead
[ɜ:] heard, learn	[i:] heat, meat

Silent or missing letters: know, honest, debt [j] in use, fuse.



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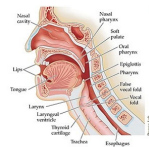


Phonetics

Phonetics tries to determine and describe the properties of sounds used in communication.

Articulatory Phonetics

- How are speech sounds produced?



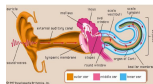
Acoustic Phonetics

- How do speech sounds "travel" from one person to another?



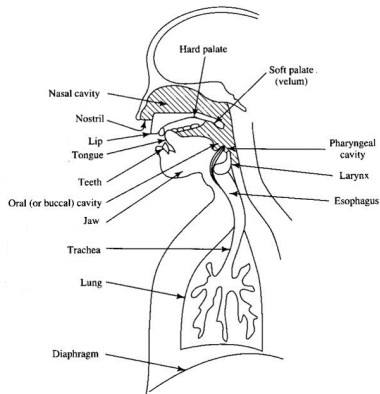
Auditory Phonetics

- How do speaker and hearer perceive these sounds?



Articulatory Phonetics

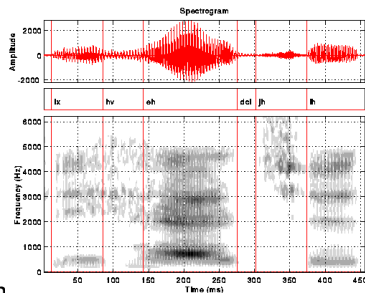
- ▶ Describes sounds with regard to the organs of speech
- ▶ Which organs of speech are involved?
- ▶ How are these organs manipulated?





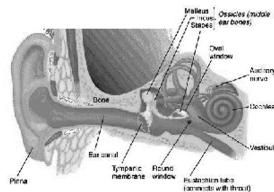
Acoustic Phonetics

- ▶ What happens to the air between speaker and listener?
- ▶ Study of the acoustic properties of speech sounds
- ▶ Use of technical devices (e.g. sonograph) to analyze the frequencies that sounds consist of.

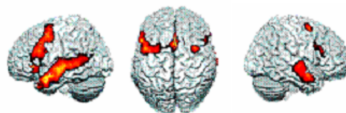




Auditory Phonetics

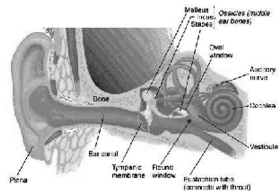


- ▶ How are speech sounds perceived by (the ear of) the speaker and the listener?
- ▶ How does the brain distinguish between sounds?
- ▶ How much do we "need to understand"?

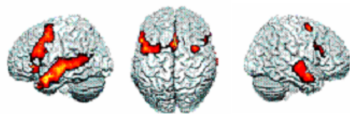




Auditory Phonetics

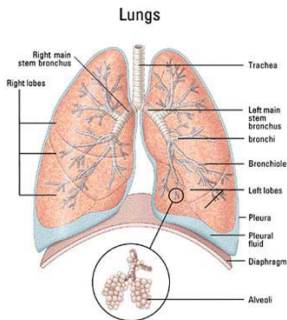


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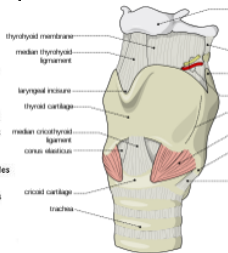


Speech Sound Production System

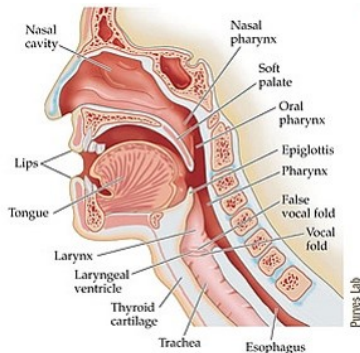
Lungs



Larynx (voice box)



Vocal tract





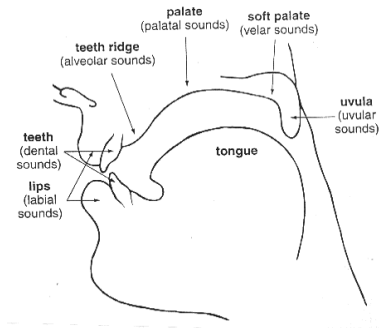
Egressive Pulmonic Airstream Mechanism

- ▶ Mechanism that produces an air stream that is pushed up from the lungs and leaves the body through the mouth or nose.
- ▶ Produces almost all speech sounds in English with the exception of sounds which we produce when we are in emotional distress (e.g. fear) or when we are counting to ourselves.

Organs of Speech

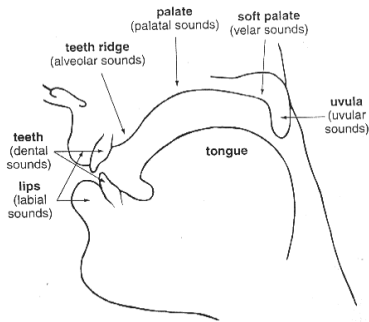
- ▶ The primary function of the organs of speech is/was in fact not speech itself, because the speech organs evolved primarily for ...

- ▶ chewing, biting, (especially teeth)
- ▶ sucking, mouth hygiene (tongue)
- ▶ to prevent us from choking on something (soft palate)
- ▶ to enable us to strain (hold the breath) when we want to lift something heavy.



Organs of Speech

- ▶ alveolar ridge (Zahndamm)
- ▶ hard palate (harter Gaumen)
- ▶ velum/soft palate (weicher Gaumen/ Gaumensegel)
- ▶ uvula (Zäpfchen)
- ▶ pharynx (Rachen)
- ▶ epiglottis (Stimmritze)
- ▶ larynx (Kehlkopf)
- ▶ vocal cords/folds (Stimmbänder)
- ▶ trachea (Luftröhre)

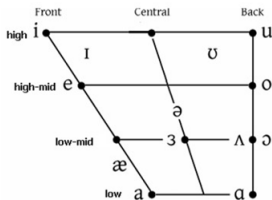




Sound Categorization



Vowels



Approximants
 Laterals
 Semivowels



Consonants

Table of English Consonants

	Bilabial	Labio-dental	Dental	Alveolar	Post-alveolar	Palatal	Velar	Glottal
Plosive	p b			t d			k g	
Affricate					tʃ dʒ			
Fricative		f v	θ ð	s z	ʃ ʒ		(x)	h
Nasal	m			n			ŋ	

Lateral				l				
Approximant	w				r	j		

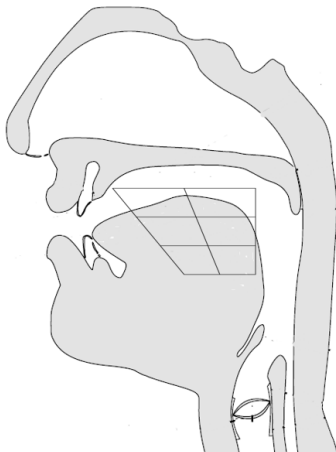


Vowels

ɪ	p <u>i</u> t	ɔː	b <u>o</u> rn
e	p <u>e</u> t	uː	b <u>oo</u> n
æ	p <u>a</u> t	aɪ	b <u>i</u> te
ʊ	p <u>o</u> t	eɪ	b <u>a</u> it
ʌ	b <u>u</u> t	ɔɪ	b <u>oy</u>
ʊ	b <u>oo</u> k	əʊ	t <u>oe</u>
ə	m <u>o</u> th <u>e</u> r	aʊ	h <u>ou</u> se
iː	b <u>ea</u> n	ʊə	p <u>oo</u> r
ɜː	b <u>ur</u> n	ɪə	<u>e</u> ar
ɑː	b <u>a</u> rn	eə	<u>a</u> ir



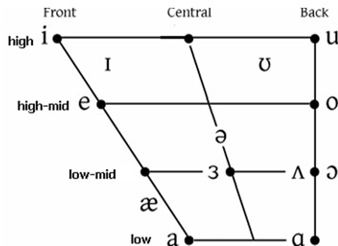
Vowels





Vowels

- ▶ Produced by ...
 - ▶ opening the lips
 - ▶ vibrating the vocal cords
 - ▶ raising the tongue to various heights.
- ▶ Vowel sounds are produced without obstruction of the air stream.
- ▶ Can form the nucleus of syllables (central, pulse-bearing part of syllables).
- ▶ Vowels are generally more sonorous than consonants.
- ▶ In English, vowel sounds are always voiced.

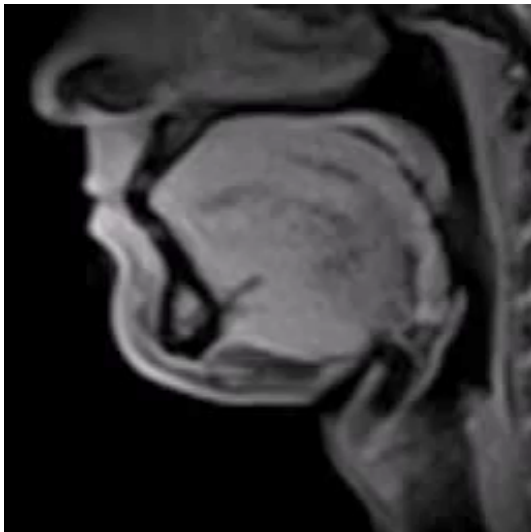




Vowels

Categorization of vowels:

- ▶ Height of the tongue:
 - ▶ high (sometimes also closed)
 - ▶ mid (sometimes also mid-closed/mid-open)
 - ▶ low (sometimes also open)
- ▶ Position of highest part of tongue:
 - ▶ front
 - ▶ central
 - ▶ back
- ▶ State of lips:
 - ▶ Rounded or unrounded (latter sometimes spread and neutral)
 - ▶ Roundedness: only affects $ʊ$, $ɔː$, and $uː$ in English.





Consonants

- Produced with obstruction of the air-stream, e.g. lip against lip [m], tongue between teeth [θ], tongue against teeth-ridge [t] and [d], tongue against soft-palate [g], etc.

Table of English Consonants								
	Bilabial	Labio-dental	Dental	Alveolar	Post-alveolar	Palatal	Velar	Glottal
Plosive	p b			t d			k g	
Affricate					tʃ dʒ			
Fricative		f v	θ ð	s z	ʃ ʒ		(x)	h
Nasal	m			n			ŋ	
Lateral				l				
Approximant	w				r	j		



Consonants

p	<u>p</u> ip	ʒ	mea <u>s</u> ure
b	<u>b</u> ib	h	<u>h</u> en
t	<u>t</u> en	tʃ	<u>ch</u> urch
d	<u>d</u> en	dʒ	<u>j</u> udge
k	<u>c</u> at	m	<u>m</u> an
g	<u>g</u> et	n	<u>n</u> ow
f	<u>f</u> ish	ŋ	<u>sing</u>
θ	<u>th</u> igh	l	<u>l</u> et
ð	<u>this</u>	r	<u>ri</u> de
s	<u>s</u> et	w	<u>w</u> et
z	<u>z</u> oo	j	<u>y</u> et
ʃ	<u>sh</u> ip	v	<u>v</u> ery

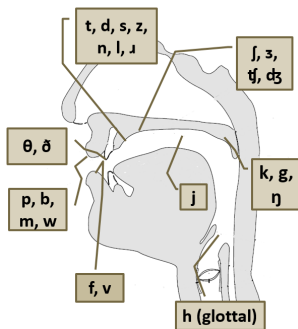


Consonant classification

- ▶ Consonants are classified according to their voicedness and according to where and how the air stream is obstructed. The relevant criteria, hence, are...
- ▶ State of glottis
 - ▶ voicedness or voicelessness (voicing)
- ▶ Place of articulation
 - ▶ bilabial, labio-dental, alveolar, glottal, etc.
- ▶ Manner of articulation
 - ▶ plosive, nasal, etc.

Consonant classification

- ▶ Soft palate/velum (velar)
k, g, ŋ
- ▶ Hard palate (palatal)
j
- ▶ Palate/alveolar (palatal-alveolar)
ʃ, ʒ, tʃ, ʤ
- ▶ Alveolar ridge (alveolar)
t, d, n, s, z, l, ɹ
- ▶ Teeth (dental)
θ, ð
- ▶ Lips (labial)
p, b, m, w (bilabial); f, v (labio-dental)





Consonant classification

- ▶ Plosives (also called stops, Verschlusslaut)
p, b, t, d, k, g
- ▶ Fricatives ("with friction", Reibelaut)
f, θ, s, z, ʃ, ʒ
- ▶ Affricates (plosives and homorganic fricatives, Affrikate)
tʃ, dʒ
(compare German: ts, pf, ʔps)
- ▶ Nasals ("through the nose", Nasale)
m, n, ŋ
- ▶ Approximants ("getting close", Näherungslaut)
w, ɹ, l, j



Transcription

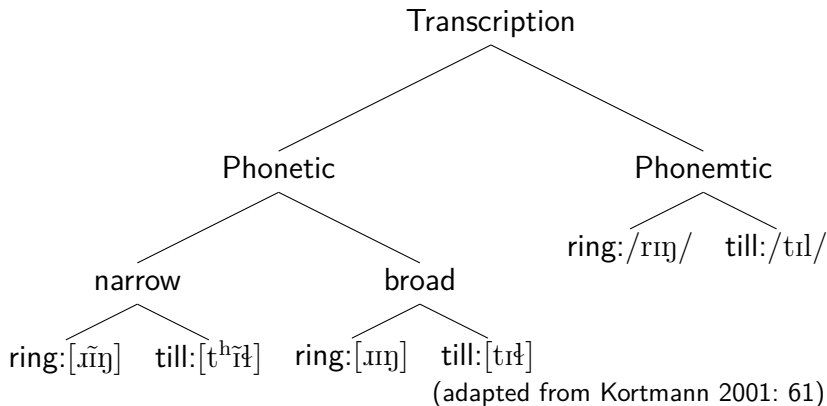




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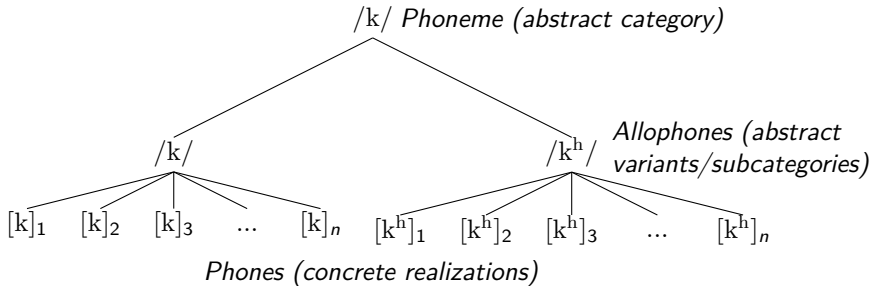
Phonology: Today's topics and terms

- ▶ Segmental and suprasegmental phonology
- ▶ Phoneme
- ▶ Minimal Pair
- ▶ Allophone
- ▶ Free variation and complementary distribution
- ▶ Syllable
- ▶ Phonological processes
- ▶ Word stress and sentence stress



Phonology: Allophony and Syllables

Functional level (Phonology)

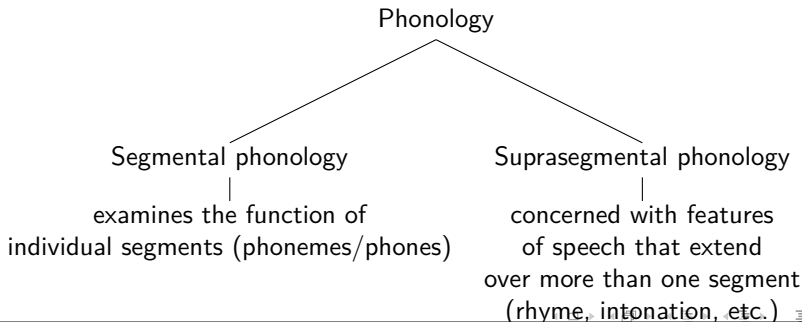


Descriptive level (Phonetics)



Phonology

- ▶ Concerned with the speakers knowledge of the sound system of languages.
- ▶ Studies the so-called sound inventory, its function and (mental) organization.
- ▶ Study of sounds on the functional level.





Phonology is NOT (so much) concerned with the actual articulation of sounds! It is concerned with the function and interaction of phonemes!



Segmental Phonology

- ▶ Concerned with the function of segments.
- ▶ Some segments distinguish meaning:

(10) /lait/ : /bait/

- ▶ If two sounds differ/are in contrast/are distinctive/are in opposition, the contrasting units are called *phonemes*.



Phonemes

- ▶ Phonemes are the smallest meaning-distinguishing units in language.
- ▶ Phonemes are not actual sounds (they are not themselves articulated) but abstract mental representations (ideal sounds) that underlie concrete articulation.
- ▶ Phonemes are identified by applying a *minimal pair test*.
- ▶ The so-called phoneme inventory of a given language encompasses all contrasting sounds that are identified by the minimal pair test.



Minimal pair test

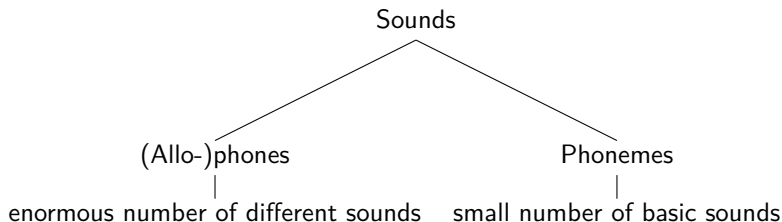
- ▶ Chose words that ...
 - ▶ have the same number of segments
 - ▶ differ in only one sound (segment)
 - ▶ differ in meaning
- ▶ /s/ and /f/ are phonemes in English because they create a difference in meaning.

Word	Segment		
	1	2	3
sun	s	ʌ	n
fun	f	ʌ	n



Every language uses a relatively small number of sounds which are combined in a relatively large number of different ways. [...] The enormous numbers of different sounds that can be found in any given language can be classified and grouped into a small number of basic sounds.

(Nathan 2008: 28)



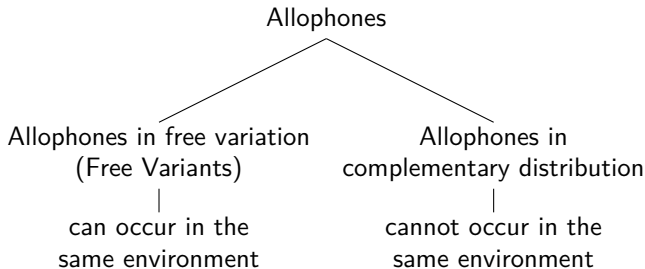


Allophones

- ▶ Allo "other" *Greek* + phone "sound" *Greek*
- ▶ Subcategories of phonemes (basic "sounds").
- ▶ Allophonic variation occurs in all languages, but the patterning of phonemes and allophones is language specific.
 - ▶ Laymen regard allophones of one phoneme as the same sound.
 - ▶ Phonetically (slightly) different.



Allophones



Word	Segment		
	1	2	3
rip	ɹ	ɪ	p
rip	ɹ	ɪ	p̚
rip	ɹ	ɪ	?p

Word	Segment		
	1	2	3
til	t	ɪ	ɫ
love	l	ʌ	v
lol	l	o	ɫ



Allophones in complementary distribution

- ▶ Allophones in complementary distribution are variants of the same phoneme that cannot occur in the same phonetic or phonological environment.
- ▶ Allophones in complementary distribution are like ...

Cannot occur in the same environment!

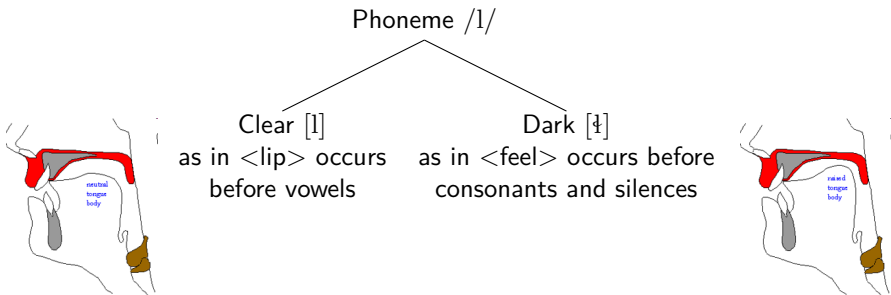
Louis Lane is in danger Daily Planet office





Allophones in complementary distribution

- ▶ Clear l and dark $ɫ$ are allophones in complementary distribution.





Phonotactic Rules

- ▶ Rules that define the translation of knowledge about language into actual speech sounds.
- ▶ Such rules are not universal, but language specific.
- ▶ In English, the voiced alveolar lateral approximant becomes velarised when it occurs before a consonant or silence:
 - ▶ phonetic form [l^ɰ] and [bɪl^ɰd]
 - ▶ phonemic form /l^ɰ/ and /bɪl^ɰd/
- ▶ In German, all word final plosives become devoiced. Thus – in terms of voicing – there is no difference between the pronunciation of <Rad> and <Rat>.



Distinctive Features

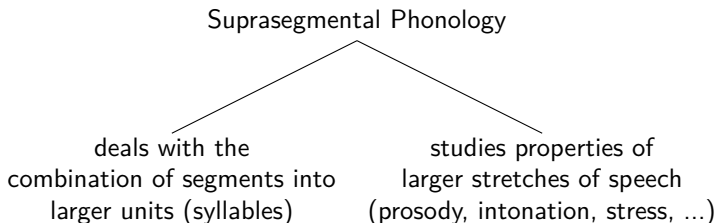
- ▶ Distinctive feature of minimal pairs [zi:t] and [si:t] or [bit] and [bid] is voicing (manner and place of articulation are identical). Sounds can be characterised by bundles of distinctive features marked by (-) or (+).

Feature	Sound			
	/b/	/d/	/g/	/k/
plosive	+	+	+	+
voiced	+	+	+	-
(bi-)labial	+	-	-	-
alveolar	-	+	-	-
velar	-	-	+	+



Suprasegmental Phonology

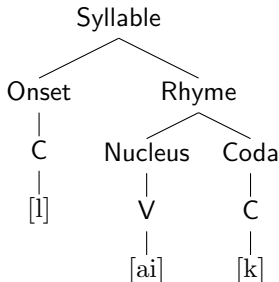
- Suprasegmental phonology deals with phonological properties that extend over more than a single segment.





Syllable

- ▶ Phonological units above phoneme level
- ▶ Smallest rhythmic unit of speech
- ▶ Monosyllabic words: words with only one syllable
- ▶ Polysyllabic words: words with at least two syllables.
- ▶ In English, all syllables contain a nucleus (or peak, core, center), a coda and an onset.





Syllable

- ▶ Nucleus
 - ▶ central element of a syllable.
 - ▶ usually a vowel sound (V).
- ▶ Coda
 - ▶ up to 4 consonants that follow the vowel (CCCC).
- ▶ Onset
 - ▶ up to 3 consonants that precede the vowel (CCC).
- ▶ Rhyme
 - ▶ nucleus and coda form the *rhyme*
- ▶ Open Syllables are syllables in which the coda is empty.
- ▶ Closed Syllables are syllables in which one or more consonants follow the nucleus.



Phonological processes

- ▶ Sounds are changed to fit their surrounding/ environment and speakers adjust the sounds they produce according to rules of their language without realizing it.
- ▶ Such adaptations are known since the 19th century and called "divergences".
- ▶ Divergences are either ...
 - ▶ physiophonetic: phonetically motivated
 - ▶ paleophonetic: conventionally or historically motivated



Phonological processes

- ▶ Progressive Assimilation
 - ▶ The noun plural marker, the singular present tense (3rd person) marker, the possessive marker, and sibilants always agree in voicing with the proceeding sound.
 - ▶ Progressive assimilation moves forward to the following segment.

Word/phrase	Transcription
dogs	[dɒg z]
cats	[kæt s]
faxes	[fæks ɪz]
he walks	[hɪgɔʊ z]
he goes	[hɪwɔ:k s]
Rick's	[rɪk s]
John's	[dʒɒn z]
Church Street	[tʃɜ:tʃtɹi:t]



Phonological processes

▶ Regressive Assimilation

- ▶ All nasals become [m] before /p/, /b/, and /m/.
- ▶ All nasals become [ŋ] before /k/ or /g/.
- ▶ Regressive assimilation moves backwards to the preceding segment.

Word/phrase	Transcription
ten bats	[tembæts]
don't be	[dəʊmpɪ]
drink!	[d.rɪŋk]

▶ Reciprocal Assimilation

- ▶ Reciprocal Assimilation occurs if two neighbouring sounds are combined to form a new sound.

Word/phrase	Transcription
don't you	[dəʊntʃu:]
kiss you	[kɪʃu:]
drink!	[d.rɪŋk]



Phonological processes

- ▶ Assimilation
 - ▶ Assimilating effect or influence on sounds by neighboring sounds.
 - ▶ Most frequent type of assimilation is regressive assimilation.
 - ▶ Tendency to produce homorganic sounds, assimilation with respect to place of articulation to increase ease of articulation
- ▶ Linguistic economy
 - ▶ The "principle of minimal effort" is mainly responsible for allophonic variation.



Phonological processes

► Deletion

- Deletion refers to cases in which sounds are omitted.
- Syncope
 - deletion in the middle of a word.
- Apocope
 - deletion at the end of a word.

Word/phrase	Full	Reduced
every	[evəri]	[evri]
family	[fæməli]	[fæmli]
lastly	[lɑːstli]	[lɑːsli]
left behind	[lefbihaɪnd]	[leftbihaɪnd]



Phonological processes

- ▶ Insertion
 - ▶ Prothesis
 - ▶ Sound is added at the beginning of a word
 - ▶ Spanish does not permit /sk/ clusters at the beginning of words.
 - ▶ School (sp.) <escuela> = [eskwelΛ]
 - ▶ Epenthesis
 - ▶ Sound is inserted in the middle of a word or phrase.
 - ▶ athlete = [æθəli:t]



Phonological processes

- ▶ Insertion
 - ▶ Excrescence
 - ▶ sound added is a consonant.
 - ▶ Anaptyxis
 - ▶ sound added is a vowel (schwa, i.e. /ə/)
 - ▶ Liaison
 - ▶ Linking r in non-rhotic dialects (if following word begins with a vowel)
 - ▶ <My car is gone> = [maɪkɑːrɪsgɒn]
 - ▶ Intrusion
 - ▶ Intrusive r between words, even if there is no <r> in the spelling.
 - ▶ <law and order> = [ləʔənɔːdə]



Stress

- ▶ ... contributes to understanding.
- ▶ ... force used to produce a syllable.
- ▶ ... indicated by cvc^1cvc .
- ▶ ... in longer polysyllabic words, we can distinguish between primary stress cvc^1cvc and secondary stress $_{,}cvcv^1cvcv$.
- ▶ ... important feature in structure not only of words, but also of phrases and sentences.

- (11) a. $[im^1pɒ^rt]$ (import; verb)
b. $[^1impɒ^rt]$ (import; noun)

- (12) a. $[en^1tra:ns]$ (entrance; verb meaning delight, enchant)
b. $[^1entrəns]$ (entrance; noun meaning door, gate)



Stress

- ▶ Word stress
 - ▶ English words with two syllables or more will have some syllable more prominent than others.
 - ▶ Due to the massive influx of foreign vocabulary, word stress is generally not predictable (rule of thumb for word stress: always stress the last but one syllable)
 - ▶ Word stress can be meaning distinguishing.

(13) a. [im'pɔrt] (import; verb)
 b. ['impɔrt] (import; noun)
- ▶ Word stress can cause vowel change and languages vary concerning whether their vowels are weakened when they appear in unstressed syllables (as in English, Russian) or not (as in German).



Stress

- ▶ Sentence stress
 - ▶ Sentence stress depends to a larger part on rhythm.
 - ▶ Rhythm refers to the distribution of stressed syllables in a phrase or sentence.
 - ▶ Lexical items that carry main semantic content (nouns, adjectives, verbs, adverbs) take stress.
 - ▶ Monosyllabic function words (grammatical PoS like prepositions, conjunctions personal pronouns) do not take stress and their vowels become schwas.



Stress

► Shift Stress (Switch stress)

- Shift stress refers to situations in which syllables with primary stress in citation form become unstressed.
- Shift stress affects words (Adj) with more than one stress in larger constructions, e.g. in NPs.
- Shift stress is typically English.
- It involves word and sentence stress.

1. Adjectives ending in -ese indicating nationalities: *Chinese, Japanese, Burmese, Portuguese* ...
e.g. *The guide is Chinese* /tʃaɪ'niz/
A Chinese guide /tʃaɪ'niz 'gaɪd/
2. Numbers above twelve: *thirteen, fifteen, twenty-two, fifty-six* ...
e.g. *Jane's thirteen* /θɜː 'tiːn/
John's twenty-two /ˌtwen ti 'tuː/
Thirteen books /θɜː tiːn 'bʊks/ *with*
twenty-two pages /ˌtwen ti tuː 'peɪ dʒɪz/ *in colour*
He died in 1313 /θɜː tiːn θɜː 'tiːn/
3. Many adjectives starting with un-: *unhappy, unarmed, unclear* ...
e.g. *She felt unhappy* /ˌʌn 'hæp i/
An unhappy feeling /ˌʌn hæp i 'fiː lɪŋ/
4. Many compound adjectives of the kind ADJ+NOUN+ed: *red-haired, blue-eyed, two-faced* ...
e.g. *They're all red-haired* /ˌred 'heəd/
A red-haired youth appeared /ˌred heəd 'juːθ/
5. Some common adverbs which can also be used adjectivally: *outside, inside, next-door, upstairs, downstairs* ...
e.g. *They live next-door* /ˌnekst 'dɔː/
Our next-door neighbour /ˌnekst dɔː 'nei bə/
6. Names of some cities and other geographical landmarks which can also be used adjectivally: *Berlin, Budapest, Hyde Park, Notting Hill* ...
e.g. *They went to Berlin yesterday* /ˌbɜː 'lɪn/
The Berlin Police have arrested him /ˌbɜː lɪn pə 'liːs/
Hyde Park /ˌhaɪd 'pɑːk/
Hyde Park Corner /ˌhaɪd pɑːk 'kɔː nəl/



Stress

- ▶ Contrastive Stress (Emphatic Stress)
 - ▶ Usually the last stressed word is most important to the speaker.
 - ▶ If two elements within a sentence are contrasted, these elements are stressed

(14) I didn't want a small piece of cake, I wanted a large piece.

- ▶ Sometimes marked by making the stress extra loud.
- ▶ Usually marked with a falling pitch, i.e. by starting the contrastive stress with higher pitch that is followed by syllables with considerably lower pitch.

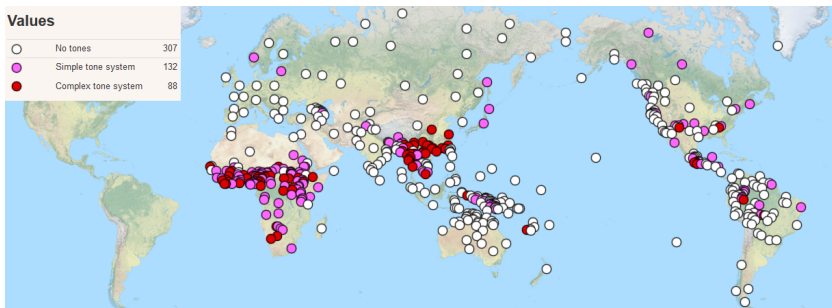


Pitch

- ▶ The faster the vocal cords vibrate, the higher the pitch.
- ▶ All voiced sounds can be produced with different pitches.
- ▶ Pitch is usually described in terms of high and low rises and falls in pitch across a stretch of speech
- ▶ In English, intonation helps marking syntactic units and vaguely corresponds to punctuation.

Pitch

- In many languages (not English) the meaning of words can differ according to pitch (such languages are called "tone languages").



World Atlas of Language Structures Online



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Morphology: Building Meaningful Blocks

- ▶ An investigation into what elements an utterance consists of is called an morphological analysis.
- ▶ In fact, morphology means "study of forms" and was adopted from biology.
- ▶ In the middle of the 19th century the term was employed to describe the study of basic meaningful elements.
- ▶ However, we do not need to look at e.g. Swahili to discover that words are made up of a number of "meaning bearing elements" ...
... but anyways, let's do that ...



Morphology: Building Meaningful Blocks

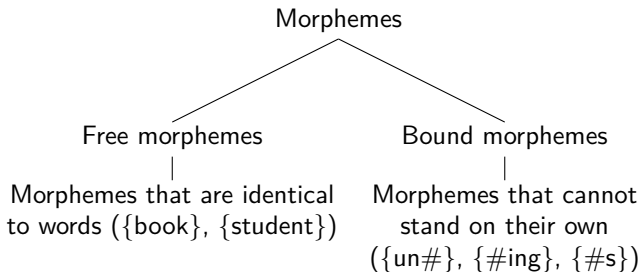
Nitakupenda. *means* I will love you.

- ▶ How many "words" does "I will love you!" contain and how many words does "nitakupenda" contain?
- ▶ English "words" and Swahili "words" are not the same but maybe both contain similar "meaning elements".
- ▶ Maybe it is better to look at these meaning elements than at words ...
- ▶ But what are these elements?



Morphology: Building Meaningful Blocks

- ▶ Smallest meaning-bearing units of language.
- ▶ Word contains at least one morpheme





Morphology: Building Meaningful Blocks

talks talker talking talked

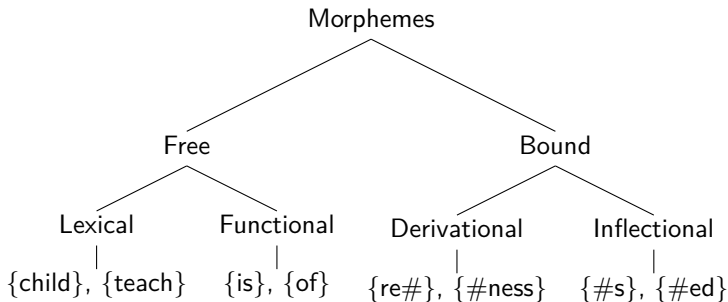
- ▶ All words above have one element in common (talk), but in all of the words some element is added ($\{\#s\}$, $\{\#er\}$, $\{\#ing\}$, $\{\#ed\}$).
- ▶ Let's look at another example ...

(15) How many morphemes is *<tourists>* made up of?

morpheme	{tour#}	{#ist}	{#s}
type	free	bound	bound
meaning	sth. to do with "tour"	person doing sth.	plural



Morphology: Building Meaningful Blocks





Types and tokens

(16) The students borrowed the books from the library.

- ▶ Word Types
 - ▶ The sentence above consists of 6 words, since "the" appears 3 times and is, thus, counted only once.
- ▶ Type frequency
 - ▶ Frequency of words with regard of repetition. If words appear repeatedly, they are only counted once.
- ▶ Word tokens
 - ▶ The sentence above consists of 8 word tokens.

Token frequency

- ▶ Word occurrence without regard of repetition. If words appear repeatedly, they counted as many times as they occur.



Open and closed classes

We can not create words however we like. Some word classes accept new members while other word classes don not.

Open-Closed class distinction

Open Class Words (adopt new members)	Closed Class Words (do not adopt new members)
Content Words (lexical; nouns, adjectives, verbs, and adverbs)	Function Words (grammatical; articles, prepositions, quantifiers, etc.)

Content-Function distinction



Affixes across the world

Language	Stem	Derivational Affix	Inflectional Affix	Full Word
English	{dark}	{#en#} (‘make’)	{#ed} (‘past’)	darkened
Aztec (Mexico)	{mic} (‘die’)	{#tia#} (‘cause to’)	{#s} (‘future’)	mictias (‘will kill’)

Kanuri (Nigeria)	Adjective	Noun
(‘excellen(t)/-ce’)	karite	nemkarite
(‘big/-ness’)	kura	nemkura
(‘small/-ness’)	gana	?
(‘bad/-ness’)	?	nemdibi



Affixes across the world

- ▶ Different languages employ different means to produce inflectional markings of forms.
- ▶ For example, in Ganda (Niger/Congo), not the suffix {#s} is employed to mark ('plural'), but the prefix {aba#}.

Ganda (Niger/Congo)	Singular	Plural
('doctor')	omuwaso	abawaso
('woman')	omukazi	abakazi
('girl')	omuwala	?
('heir')	?	abasika



Affixes across the world

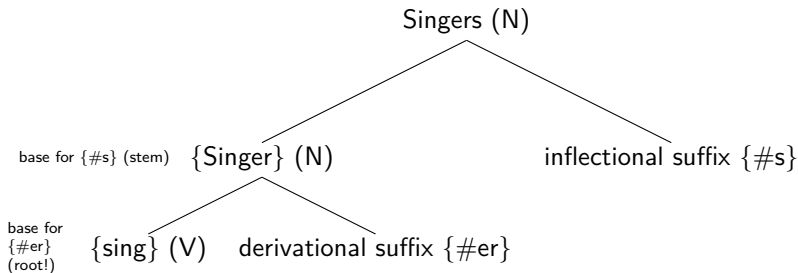
Ilocano (Philippines)	Singular	Plural
('head')	úló	ulúlo
('road')	dálan	daldálan
('life')	bíag	bibíag
('plant')	múla	mumúla

- ▶ In Tagalog, to create the imperative of an infinitive, insert the infix {#um#}.
- ▶ To create the future from an infinitive, the first syllable has to be reduplicated.

Tagalog (Philippines)	Infinitive	Imperative	Future
('read')	basa	bumasa	babasa
('call')	tawag	tumawag	?
('write')	sulat	?	susulat



- ▶ Base (Stem)
 - ▶ Any form to which an affix is attached
- ▶ Root
 - ▶ Single morphemes that cannot be analyzed any further (morphologically)





Affixes

Besides prefixes and suffixes, there are two/three more types of affixes.

- ▶ Infixation
 - ▶ Infixes are inserted into the base (abso-blooming-lutely)
- ▶ Circumfixes
 - ▶ Circumfixes are attached to beginning and end of the base (ge-sag-t or down-load-ed)
- ▶ Zero-morphs
 - ▶ Some linguists argue that the plural morpheme in "information" or "sheep" exists but it is not realized.



Allomorphy (Morphological Alternation)

Problem: One morpheme may be realized differently under different conditions.

► Indefinite and definite article

(17) /ə/ question : /ən/ answer

/ə/ book : /ən/ author

/ə/ fence : /ən/ idea

In isolation /ei/

(18) /ðə/ question : /ðI/ answer

/ðə/ book : /ðI/ author

/ðə/ fence : /ðI/ idea

In isolation /ðI/

► Different variants = allomorphs



Allomorphy (Morphological Alternation)

Types of Allomorphy

- ▶ Phonological Allomorphy

(19) {#ed} = /d/, /t/, /ɪt/

- ▶ Weak suppletion (weak suppletive allomorphy)

(20) /aɪ/ (buy) : /ɔː/ (bought)

- ▶ Phonological similarity but no phonological rule/condition.
 - ▶ Strong suppletion (strong suppletive allomorphy)
- (21) go : went / be : is : are : was : were
- ▶ No phonological similarity and no phonological rule/condition



Allomorphy (Morphological Alternation)

Determinants of Allomorphy (Morphological Alternation)

- ▶ Phonological Conditioning

(22) /hɔːɪz/ : /kæts/ : /dægz/

- ▶ Choice of allomorph is determined by phonological environment

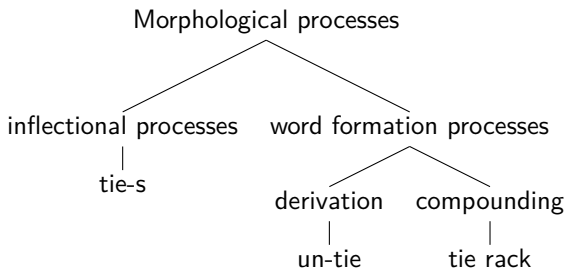
- ▶ Morphological Conditioning

(23) Gen. wife's /waɪfs/ : Pl. wives /waɪvz/

- ▶ Morphological context determines choice of allomorph

- ▶ Lexical Conditioning

(24) /baɪ/ : /bɔːt/ vs /krai/ : /kaɪd/ (not /krɔːt/)





Word Formation

Word formation

- ▶ Most Productive Word Formation Processes
- ▶ Derivation
 - ▶ Derivation means that new lexemes are formed by adding an affix, usually prefixes or suffixes, that may cause the word to change its word class

$$(25) \quad \{\text{cool}\}_{Adj} + \{\# \text{ness}\} = \{\text{coolness}\}_N$$

- ▶ Compounding
 - ▶ Combining at least 2 existing words are to form a new one.

$$(26) \quad \text{"wheel"} + \text{"chair"} = \text{"wheel chair"}$$

(In English, compounds are commonly not orthographically combined).



Word Formation

Derivation

► Derivational Prefixes

- ... modify the meaning of words without any changes regarding the lexical class.
- ... are often Latin or Greek origin ($\{\text{pre}\# \}$, $\{\text{meta}\# \}$, $\{\text{pro}\# \}$, ...).

► Derivational Suffixes

- ... may change the lexical class of words

$$(27) \quad \{\# \text{er}\} + \{\text{read}\}_V = \text{reader}_N$$

- ... may produce new words with different meanings.

$$(28) \quad \{\text{interview}\} + \{\# \text{er}\} : \{\text{interview}\} + \{\# \text{ee}\}$$

$$(29) \quad \{\text{un}\# \} + \{\text{kind}\} = \{\text{unkind}\}$$



Word Formation

Conversion (Zero Derivation)

- ▶ A word comes to belong to another word class without addition of an affix.
- ▶ Many English words exist as nouns and verbs

(30) smell_V (+ { $\emptyset$ }) : smell_N

(31) taste_V (+ { $\emptyset$ }) : taste_N

(32) talk_V (+ { $\emptyset$ }) : talk_N



Word Formation

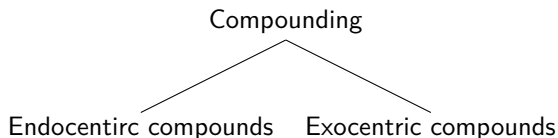
Compounding

- ▶ Combining at least two existing words to form a new word.
- ▶ Most compounds in English are nouns, verbs and adjectives (exceptions: "into", "onto", ...).
- ▶ Usually the head of a compound (right-hand element) determines its lexical class.
- ▶ English compounds may be written as one word, with or without hyphen, or as two words.



Word Formation

Compounding



- ▶ Endocentric Compounds

- ▶ Meaning can be guessed from its components.
- ▶ Head is specified by the left-hand-side element.

(33) <morphology book> is a special kind of <book>.

- ▶ Exocentric Compounds

- ▶ Meaning cannot be guessed from its components.

(34) <redneck> is a US social group, not a "red" neck.

(35) <pickpocket> is not a special kind of <pocket>, but a thief.



Word Formation

Conjunction Test

- ▶ To distinguish compounds from non-compound combinations of adjectives and nouns we can use the so-called "conjunction test".
 - ▶ Insertion of another adjective into the phrase.
 - ▶ The insertion is not possible in case the phrase is a compound.
- ▶ Another method to distinguish compounds from non-compounds is to look at stress
- ▶ In compounds, stress is usually on the first element ('redneck)
- ▶ In non-compounds, stress is usually on the second element (red'neck)



Word Formation

Blending

- Blends combine non-morphemic parts of words.

(36) {breakfast} + {lunch} → {brunch}

Clipping

- Creation of new words by shortening existing ones.

(37) {professor} → {prof}

Back-Formation

- Real or supposed affix is removed to create a new word.

(38) editor → to edit; baby sitter → to baby sit



Word Formation

Coinage

- Invention of new words (word manufacture)

(39) brand-names (Weetabix)

(40) fictional characters (Hobbit)

Initialisms

- formed from the initial letters of a set of other words.

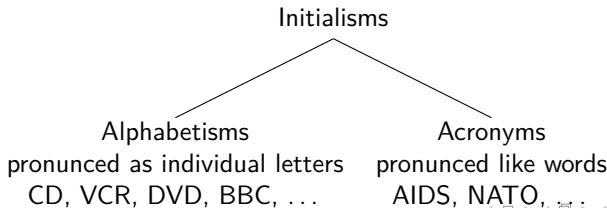




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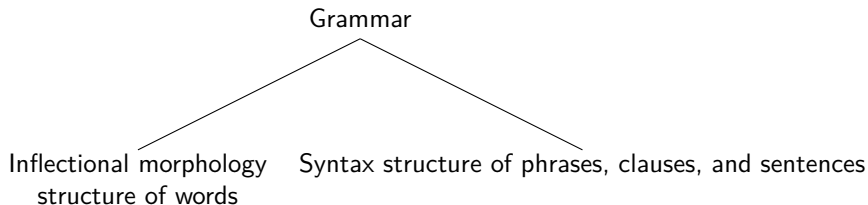
Syntax: Word Classes and Phrases

Grammar

- ▶ Study of the rule-based structure (the skeleton) of a language.
- ▶ Descriptive approach
 - ▶ ...studying language as it is actually spoken.
 - ▶ ...without criticism of colloquial features.
- ▶ Prescriptive approach (normative approach)
 - ▶ ...describes how a language should be spoken.
 - ▶ ...common among grammarians of the 18th and 19th century to lay down arbitrary rules for the correct or educated use of English.
 - ▶ No "split infinitive" (to quickly go); "I will" instead of "I shall" to indicate future tense.



Grammar and Syntax





Inflectional morphology

We have now encountered two basic ways to deal with phrases or sentences.

Example: "the lucky boys" : /ðəlʌkɪbɔɪz/

- ▶ phrases or sentences can be described in terms of sequences of sounds that we can describe and classify (phonetics and phonology):
/ð/ = voiced bilabial fricative, /ə/ = mid central unrounded vowel, ...
- ▶ phrases and sentences can be described on the morphological level:
{the} = functional, free morpheme; {luck} = lexical, free morpheme; ...



Languages in comparison

Sequencing is not unlimited, i.e. the number of grammatically accepted combinations is limited.

- Some languages achieve via inflection (Latin) what other languages achieve syntactically (English).

(41) The boy sees the girl. $\neq$ The girl sees the boy.

(42) Puellum videt puella. = Puella puellum videt. = Puellum puella videt. = Videt Puellum puella.

- In English the grammatical information (subject vs. object) is encoded in the word order, while the same information is encoded in case marking in Latin (nominative = subject; accusative = direct object).
- Languages are categorized according to how they encode grammatical information.



Languages in comparison

Agglutinating languages

- ▶ ... each grammatical morph carries exactly one piece of information.

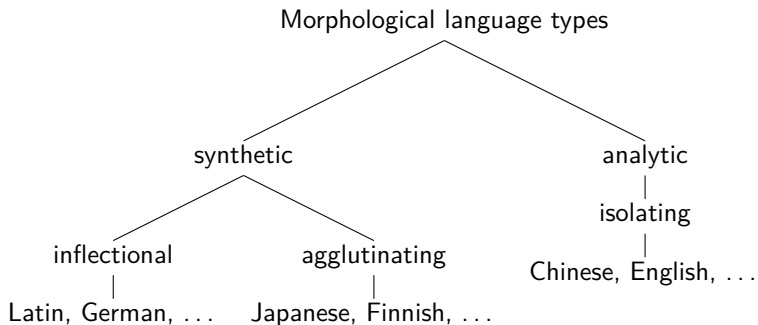
Inflectional languages

- ▶ ... one grammatical morph usually carries more than one meaning.
- ▶ For example, the Latin inflectional suffix {#us} carries the meaning(s) SG, MASC and NOM. In agglutinating languages each of the pieces of information is encoded in a different morph.

Languages are categorized according to how they encode grammatical information.



Languages in comparison





Languages in comparison

European languages

- ▶ ... have undergone a change from synthetic to analytic (this might actually be a circular process).
- ▶ ... show a decrease in the number of inflectional morphemes.
- ▶ English has lost almost all inflectional morphemes (English has been referred to as the "language of largely invariable words").



Inflectional morphology

- ▶ Of all inflectional morphemes that were present in Old English, only 8 inflectional suffixes have survived.
- ▶ The main task of inflectional morphemes is agreement.

nouns	declension	{PLURAL}:{#s} {GENITIV}:{#s}	two boys <u>s</u> the boy's <u>s</u>
verbs	conjugation	{3SG. IND. PRES}:{#s} {PAST}:{#ed} {PRES. PART}:{#ing} {PAST. PART}:{#ed}	he works <u>s</u> he worked <u>ed</u> he is work <u>ing</u> he has work <u>ed</u>
adjective	comparison	{COMPARATIVE}:{#er} {SUPERLATIVE}:{#est}	stronger <u>er</u> strongest <u>est</u>



Parts of Speech

- ▶ The grammatical vocabulary to describe "words" dates back to early descriptions of Latin and Greek (also to a minor extent Sanskrit).
- ▶ Those grammatical descriptions were quite elaborate and thus existing terms were adopted to apply in modern grammatical descriptions of late languages.

(43) The lucky boys found a backpack in
 Article Adjective Noun Verb Article Noun Preposition
 the park and they opened it carefully.
 Article Noun Conjunction Pronoun Verb Pronoun Adverb



Parts of Speech

POS	Example(s)
Nouns	boy, dogs, school, roughness, love
Articles	a, an, the
Adjectives	happy, large, strange
Verbs	has, talk, go
Adverbs	slowly, beautifully
Prepositions	at, in, on, under, near, with, without
Pronouns	I, you, he/she/it, herself, they
Conjunctions	and, or, because, when



Parts of Speech

There are basically 3 criteria to classify lexemes as nouns, verbs, adjectives, etc.

Meaning

- ▶ Refers to the traditional semantic notion of the word classes. For example, all words which name persons, objects and places are nouns.

Inflection

- ▶ Refers to the morphological properties of a word, such as plural and possessive forms of a noun.

Distribution

- ▶ Depends on the syntactic properties of a word, such as its potential positions / functions within a phrase, clause or sentence.



Parts of Speech

The classification of words (lexemes) into different syntactic categories can be problematic, because

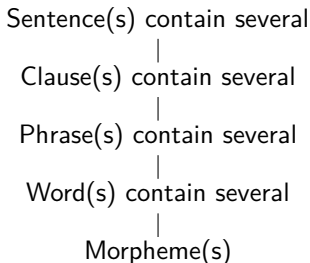
- ▶ ... some word classes are more heterogeneous than others. Due to their various modifying functions, adverbs have often been the "waste bin" for lexemes that could not be clearly assigned to other parts of speech.
- ▶ ... transitions and fuzzy boundaries between different categories.
- ▶ ... one word form can belong to more than one word class multiple classifications are possible.
- ▶ ... parts of speech are heterogeneous in themselves not all members of a certain word class exhibit all its characteristics.



Syntax

Greek: syntaxis = order, arrangement.

- ▶ Syntax is the study of the rules that enable us to recognize and generate an unlimited number phrases, clauses and sentences.
- ▶ Smaller building blocks are formed of so-called "constituents".





Phrases

The young linguist will sell her book to a colleague very soon.

- ▶ Some parts of speech belong together more closely than others.
- ▶ For example, the words <The young linguist> are more closely related than the words in the sequence <books to a>. Groups of words that belong together are called *phrases*.
- ▶ In most phrases one central element (head) is extended by adding one or several modifying elements (modifiers).



Phrases

The young linguist will sell her book to a colleague very soon.

Constituency tests

- ▶ Substitution (with a pro-form)
<The young linguist> passes as a phrase, because it can be substituted by <she>, while <books to a> fails this test.
- ▶ Questioning
Identifying phrases by asking questions that will yield this group of words as their answer.
- ▶ Coordination
Identifying phrases by coordinating them with elements of the same category or elements with a similar structure.



Phrases

The young linguist will sell her book to a colleague very soon.

Constituency tests

- ▶ Deletion

Optional phrases may be deleted without the sentence becoming ungrammatical.

- ▶ Movement

Some constituents may be moved to a different position within the sentence without the sentence becoming ungrammatical.



Phrase Structure Rules

Sequencing is not unlimited, i.e. the number of grammatically accepted combinations is limited.

- ▶ For example, the sequence *<lucky boys the> would not be rated as grammatically correct by native speakers of English.
- ▶ In English, the article <the> must stand before the adjective <lucky> which must stand before the noun <boys> (article + adjective + noun).
- ▶ Such rules are called "Phrase Structure Rules" (PSRs)
- ▶ In general, PSRs in English are more strict than PSRs in German!
- ▶ Different languages have different PSRs!



Phrase Structure Rules

Imagine you come across a language (which only consists of 3 sentences) and want to describe its grammar.

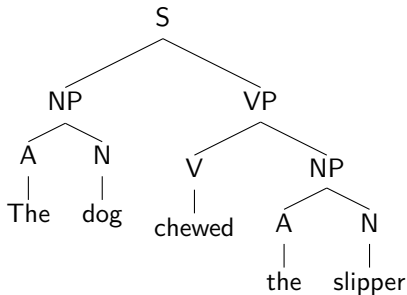
(44) The dog chewed the slipper.

(45) The woman ate the cake.

(46) A fish swam a mile.

The PSRs that can be inferred are:

1. $S \rightarrow NP VP$
2. $VP \rightarrow V NP$
3. $NP \rightarrow A N$



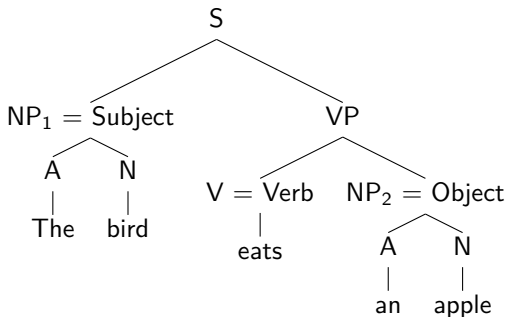
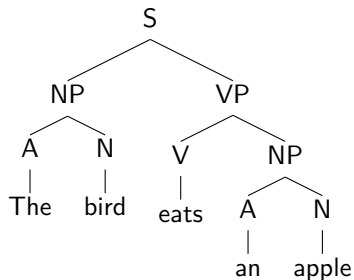


Form and Function

- ▶ So far: formal approach towards the analysis of words, phrases and clauses.
- ▶ Difference between FORM and FUNCTION
- ▶ NP and VP (also PoS tags) refer to the form of constituents.
- ▶ Subject, verb and object refer to the function of constituents.



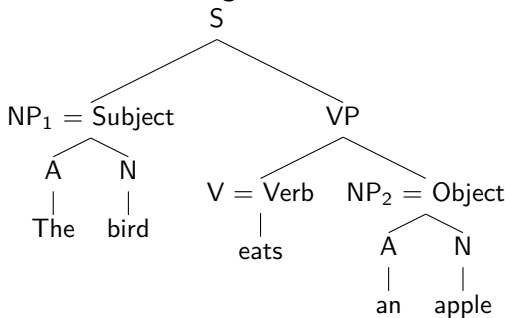
Form and Function





Form and Function

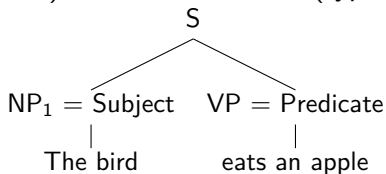
- Subjects are daughters of S and sisters of the VP.
- Objects are NPs and daughters of the VP and sisters of V.





Theme and Rheme

- Sentences can be divided into the Subject (typically a NP) and the Predicate (typically a VP).

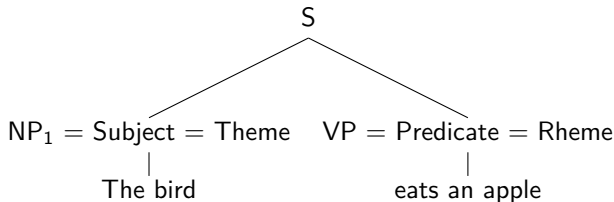


- Subjects typically represent given/known information.
- Predicates typically represent new/unknown information.



Theme and Rheme

- ▶ Theme
 - ▶ The theme refers to what is known or given (old information ~ subject).
- ▶ Rheme
 - ▶ The rheme typically represents new information (the motivation for the sentence ~ predicate: everything that is daughter to the VP).

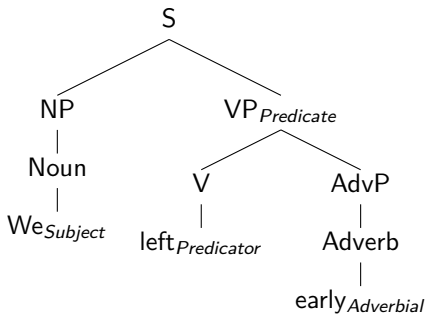




Adverbs and adverbials

Adverb $\neq$ Adverbial

- ▶ Adverb: PoS (form)
- ▶ Adverbial: functional constituent (function)

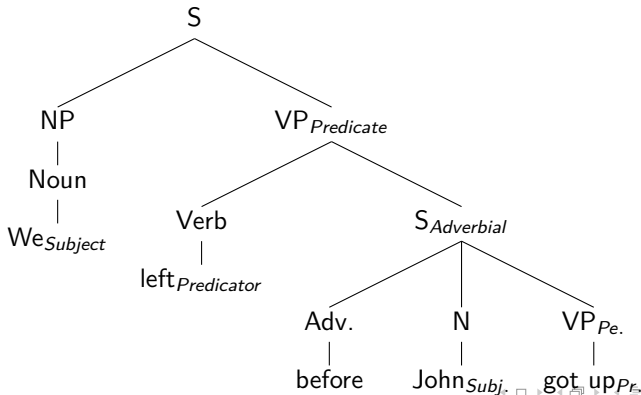




Adverbs and adverbials

Adverb $\neq$ Adverbial

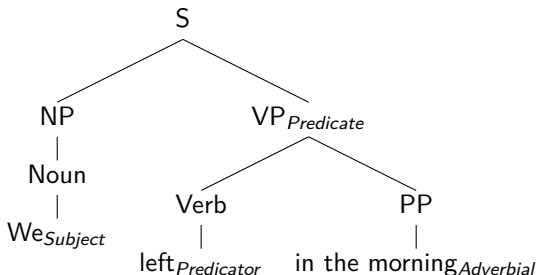
- ▶ Adverb: PoS (form)
- ▶ Adverbial: functional constituent (function)



Adverbs and adverbials

Adverb $\neq$ Adverbial

- ▶ Adverb: PoS (form)
- ▶ Adverbial: functional constituent (function)





Adverbials and subordinate clauses

Type of clause	Example
subject clause	<u>That you are here</u> is a miracle.
object clause	We knew <u>(that) he was a lousy driver</u> .
complement clause	The problem is <u>how to stay away from trouble</u> .
adverbial of time	He left <u>as soon as we had finished breakfast</u> .
adverbial of place	He waited <u>where I left him</u> .
adverbial of manner	She behaves <u>as if she has problems</u> .
adverbial of condition	<u>If you leave now</u> , you will still reach the train.
adverbial of cause	I was angry <u>because he was late</u> .
adverbial of concession	<u>Although I love good food</u> , I eat very little.
adverbial of purpose	He came <u>(in order) to help me</u> .
adverbial participle	<u>Walking along the river</u> , he watched the fishermen.

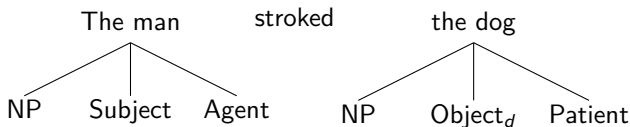


Semantic roles

Constituents (noun phrase, verb phrase, etc.) have

- ▶ syntactic relations (subject, predicate, object, etc.)
- ▶ semantic roles (agent, patient, etc.).

(47) The man stroked the dog.





Sentence types

- ▶ The sentence is the largest independent syntactic unit of a language which is not embedded in any larger syntactic construction.
- ▶ Sentences that consist out of one subject-predicate structure are called *simple sentences*.
- ▶ Sentences with more than one clause are either called ...
 - ▶ compound sentences (combination of main clauses).
 - ▶ complex sentences (combination of one main clause and at least one subordinate clause).



Sentence types

- ▶ Declarative sentences (to inform someone of sth.)

(48) Anna is singing.

- ▶ Interrogative sentences (to get information about sth.)

(49) Is Anna singing?

- ▶ Imperative sentences (to get someone do sth.)

(50) Sing!

- ▶ Exclamatory sentences
(to express our attitude towards about sth.)

(51) How beautiful Anna is singing!



Sentence Constituents

- ▶ Adjuncts are never obligatory!
- ▶ Adverbials can but do not have to be obligatory!
- ▶ Complements are necessary and can take various forms (they do not have to be NPs)



Sentence patterns

Sentences consists minimally of a subject and a predicate (other elements are objects, adverbials and complements).

Pattern	Subject	Predicate/verb	Object(s)	Complement	Adverbial
SV	The girl	was sleeping			
SVO	Her mother	was dressing	the baby (O_d)		
SVC	Little James	seemed		very happy (C_S)	
SVA	He	was sitting			on the table
SVOO	Mrs Bates	gave	her children (O_i) all her love (O_d)		
SCOC	Most people	considered	her (O_d)	a perfect mother (C_O)	
SVOA	She	has spent	all her life (O_d)		in the village



Verb types

Verbs can be divided into ...

- ▶ lexical verbs (main verbs)
- ▶ grammatical verbs (auxiliaries)

In the course of the history of the English language, the division between these two kinds of verbs has become more and more strict, so that, nowadays, English auxiliaries form a separate group which, morphologically as well as syntactically, is very different from main verbs.

- ▶ Auxiliaries can be divided into ...
 - ▶ primary verbs (*be*, *have* and *do*)
 - ▶ modal verbs (e.g. *dare*, *need* and *used to*)
- ▶ The difference between primary verbs and modal verbs is that only primary verbs may also be used as main verbs.



Verb types

Verb type	description	Sentence pattern
Intransitive verbs	require only one argument (subject)	SV (monovalent)
Prepositional verbs	require, apart from the subject, a prepositional phrase, but they do not require a direct object	SVPP (SVA)
Transitive verbs	require a subject and an object	SVO _d (can be passivized)
Intensive verbs	require a subject and a subject compliment, but they do not require an object	SVC _s
Ditransitive verbs	require a subject, an indirect object, and a direct object	SVO _i O _d
Complex-transitive	verbs require a subject, an object, and something else	SVO _d C _o or A



Verb types

- ▶ The main verb (head of VP) is central to the sentence structure
- ▶ Seven sentence patterns can be derived from different verb types.

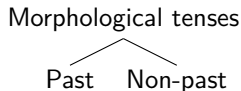
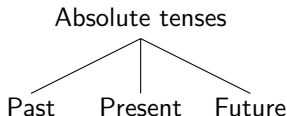
Agruments	Valency	Transitivity	Examples	Pattern
0	avalent	—	rain, snow, freeze	SV
1	monovalent	intransitive	sleep, sit, walk	SV
2	divalent	—(copula)	be, become	SVC
2	divalent	—	live, stay, last	SVA
2	divalent	monotransitive	read, take, build	SVO
3	trivalent	ditransitive	give, offer, pass	SVOO
3	trivalent	compl. trans.	consider, call	SVOC
3	trivalent	compl. trans.	put, hide, spend	SVOA



Verbs and time

Tense

- ▶ Deictic category (from Greek *deiknym* = to show):
 - ▶ depends on the moment of utterance.
 - ▶ If tense is defined by inflectional marking, English has two tenses: past (walk-ed) and non-past (walk).
 - ▶ If tense is not defined by the existence of inflectional marking, English has three tenses to place an event in one of the three possible time spheres.

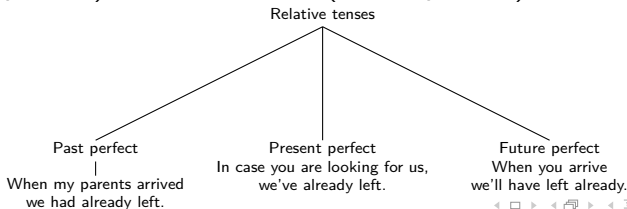




Verbs and time

Perfect (tense(s))

- ▶ Perfect (tenses) are not indexical, because they are used to locate an event in time not with reference to the moment of utterance, but with reference to some other event!
- ▶ The perfect tenses are used to express anteriority to some event in the past (Past perfect), the present (present perfect) or in the future (future perfect).



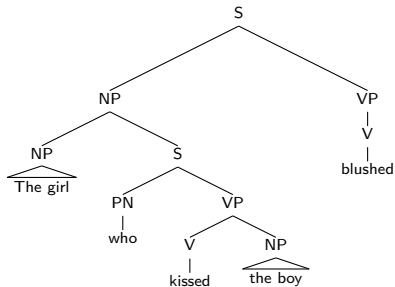
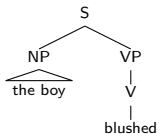


Segmentation

- (52) The boy blushed.
- (53) The girl who kissed the boy blushed.
- (54) *The girl who kissed (the boy blushed).
- (55) The girl (who kissed the boy) blushed.



Segmentation





Congruence

(56) The clown was amused.

(57) The boys who watched the clown were amused.



Sentence structure

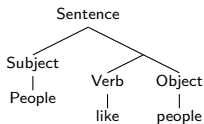
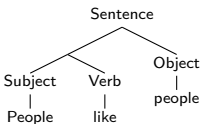
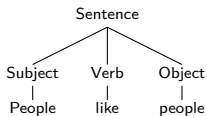




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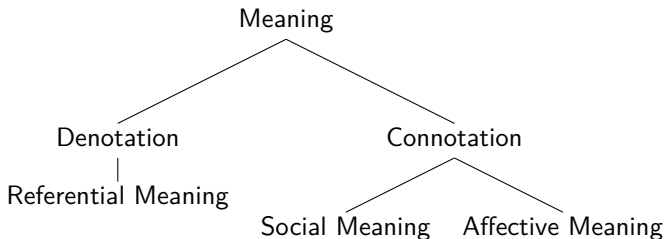
Semantics

- ▶ Semantics
 - ▶ ... concerned with referential meaning (denotation).
 - ▶ ... not concerned with what is intended by the speaker.
- ▶ Types of meaning
 - ▶ Referential meaning
 - ▶ Referent: thing in the world referred to by a word
 - ▶ Sense: idea/concept referred to by a word
 - ▶ Words always have a sense but they do not necessarily have a referent (bigfoot)
 - ▶ Social Meaning
 - ▶ Information about the speaker's background (lift vs elevator)
 - ▶ Affective meaning
 - ▶ Information about the speaker's attitude/stance toward sth. (terrorist vs freedom fighter)



Semantics

- ▶ Semantics
 - ▶ ... concerned with referential meaning (denotation).
 - ▶ ... not concerned with what words connote.





Semantic analysis

- ▶ Two pairs of terms are especially important:

Connotation	—	Denotation
Sense	—	Reference

- ▶ The above pairs are useful in explaining the organization of our mental lexicon (how words and meaning are represented and processed in our mind):
 - ▶ <Connotation>, <Sense>, and <Intension> are related to the language-internal/ intra-linguistic side of meaning.
 - ▶ <Denotation>, <Reference>, and <Extension> are related to language-external/ extra-linguistic reality.



Semantic analysis

- ▶ Connotation
 - ▶ Connotations are associations that are evoked by a certain word.
 - ▶ The word in question is defined within the network of words that we think of when we hear that word.
 - ▶ For example, the term "spring" has the connotations blossoms, green meadows, sunny weather, etc.
- ▶ Denotation
 - ▶ Denotation refers to the accurate description of terms.
 - ▶ Denotations are "definitions" of terms.
 - ▶ The denotation of spring is "Spring is one of the four seasons and follows winter but precedes summer".



Semantic analysis

► Sense

- Sense refers to the meaning of an expression within a language and is defined by its relation to other expressions.
- For example, the sense of the term cow is its linguistic description, e.g. "a large four-legged animal kept on farms to produce milk or beef".

► Reference

- Reference refers to the relationship between a linguistic expression and the person, object, entity or state of affair in the real world to which it refers.
- For example, the reference of the term cow are all the cows out there in the world.



Semantic analysis

- ▶ Feature-Based Semantics
 - ▶ Features are a set of semantic properties which define a term or expression.
 - ▶ For example, the term bird evokes semantic features like "animate" and "not human", "has feathers" and "has wings".
 - ▶ The semantic features of the term bird are [+animate], [-human], [+wings], and [+feathers].
 - ▶ When terms are described with regard to their semantic properties, they are broken down into semantic components
 - ▶ This approach is called componential analysis.



Branches of Semantics

- ▶ Lexical Semantics
 - ▶ ...concerned with the meaning of words
 - ▶ ...the relationships between the meaning of words.
- ▶ Sentential Semantics (or Phrasal Semantics)
 - ▶ ...deals with meaning of larger syntactic units larger than words, i.e. phrases, clauses and sentences
 - ▶ ...the relationship between them.



Lexical semantics - Sense Relations

- ▶ Meaning relations among words
 - ▶ Words of a language can be related in a number of ways

Relationship	Example
Synonymy	buy - purchase
Antonymy	boy - girl
Homophony	right - write
Polysemy	back: the back of the house — back: to go back
Hyponymy	fruit - apple
Meronymy	face - nose



Meaning relations among words

- ▶ Synonymy
 - ▶ Synonymy describes an extensive semantic similarity (semantic relation between words).
 - ▶ Synonyms are words with the same or nearly the same meaning (stupid – dumb; movie – film – motion picture – flick).
 - ▶ Dictionaries in which words with similar meanings are grouped together are called thesauri. A thesaurus contains lists of synonyms.



Meaning relations among words

- ▶ Antonymy
 - ▶ Opposition of meaning.
 - ▶ Words with opposite meanings are called antonyms.
 - ▶ Antonyms are opposites with respect to at least one component of their meaning, BUT share other aspects of their meaning.

Types of antonyms

Type	Example
Complementary pairs (Antonyms)	on – off
Gradable Pairs	cold – warm
Relational Opposites	employer – employee
Directional Opposites	arrive – depart



Meaning relations among words

- ▶ Antonymy and word formation
 - ▶ Concerning word formation, there are several ways to create antonyms.
 - ▶ Adding the suffixes {#ee} or {#er} (interviewee – interviewer).
 - ▶ Adding prefixes like {un#}, {in#}, {non#}, {dis#}, or {mis#} (un–able, in–sane, non–sexist, dis–honest).



Meaning relations among words

- ▶ Polysemy
 - ▶ Words that have the same pronunciation
 - ▶ Words that differ in meaning
 - ▶ The different meanings are historically or semantically related.
 - ▶ paper (something to write on) — paper (scientific publication)
- ▶ Homophony
 - ▶ Words that are pronounced the same.
 - ▶ The meaning of the words differ.
 - ▶ The individual meanings of the sound sequences are not related (right /rait/ and write /rait/).



Meaning relations among words

- ▶ Homography
 - ▶ Words that are spelled the same, but are not pronounced the same are called homographs.
 - ▶ dove /dʌv/ (bird) - dove /doʊv/ (past tense of dive)
- ▶ Homonymy
 - ▶ The term homonymy is oftentimes used synonymously with homophony but it actually refers to words that are identically pronounced and written.
 - ▶ bank /bænk/ (of a river) – bank /bænk/ (financial institution)
 - ▶ race (competition) – race (population group)



Meaning relations among words

► Hyperonymy

- The meaning of one word includes in the meaning of other words.
- For example, <fruit> is a hyperonym, because it is super-ordinate to the more specific terms <peach>, <apple>, and <orange>.

► Hyponymy

- The meaning of one word is included in the meaning of another.
- For example, <peach>, <apple>, and <orange> are hyponyms of the more general expression <fruit>.
- Also, <peach>, <apple>, and <orange> are co-hyponyms of the hyperonym <fruit>.

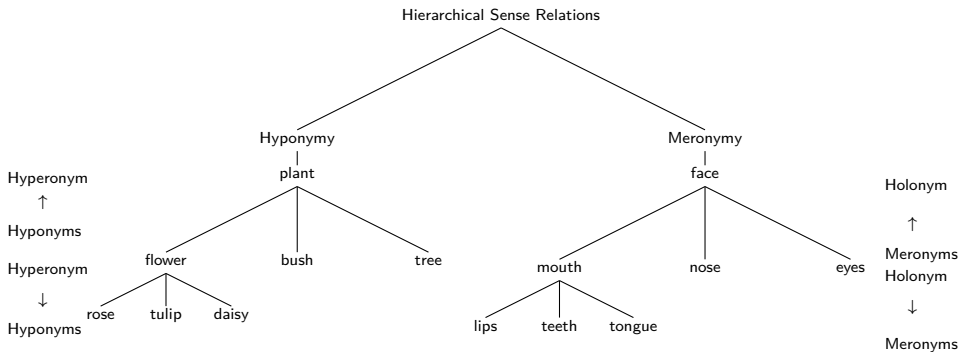


Meaning relations among words

- ▶ Heteronymy
 - ▶ The relation between different hyponyms is called heteronymy (incompatibility), because if one is true, the other cannot be true at the same time.
 - ▶ For example, if the proposition "X is a turnip" is true, the proposition "X is a rose" must be wrong.
- ▶ Meronymy
 - ▶ Meronymy refers to terms of parts of objects.
 - ▶ The difference between meronyms and hyponyms is that hyponyms refer to a hierarchy and not a part–whole relationship.
 - ▶ For example, <branches> and <trunk> are meronyms of <tree> or <nose> and <lips> are meronyms for <face>.



Meaning relations among words





Sentential semantics

- ▶ "Sentential Semantics" (Phrasal Semantics) is the study of the meaning of phrases, clauses, and sentences.
- ▶ Sentential semantics is based on the "Principle of Compositionality":

The meaning of a phrase or a sentence is determined by the meaning of its component parts and the way they are combined structurally.



Sentential semantics

- ▶ Formal Semantics
 - ▶ The type of semantics that approaches meaning by employing the notion of truth is called formal semantics.
 - ▶ The part of the meaning of a sentence that can be said to be either true or false is called proposition.
 - ▶ A proposition is true if, and only if the stated proposition is the case.
 - ▶ Truth conditions determine when a sentence can be judged as being true.



Sentential semantics

- ▶ Sentences have meanings that can be analyzed in terms of their relations to the meaning of other sentences.
- ▶ There are basically three types of semantic relations between sentences:
 - ▶ Paraphrase
 - ▶ Entailment
 - ▶ Contradiction



Paraphrases

- ▶ Sentences that have the same meaning are said to be paraphrases of each other.
- ▶ To produce paraphrases it is common so replace one word with a synonymous expression or to rephrase an active sentence in the passive voice or vice versa.
- ▶ Sentences with the same meaning cannot be true unless both sentences are true.
- ▶ Pairs of sentences that are true under the same circumstances are said to have the same truth conditions.



Entailment

- ▶ When the being true of one sentence implies the truth of another sentence their relation is referred to as entailment.

(58) Anna likes every single kind of fruit.

(59) Anna likes oranges.

- ▶ Paraphrases have the same truth conditions and entail each other symmetrically, while the entailment of sentences (34) and (35) is asymmetrical, because (34) entails that (35) is true but not vice versa.



Contradiction

- ▶ Contradiction (negative entailment) means that the truth of one sentence implies the falseness of the other.
- ▶ If the one sentence is true, the other is necessarily false.

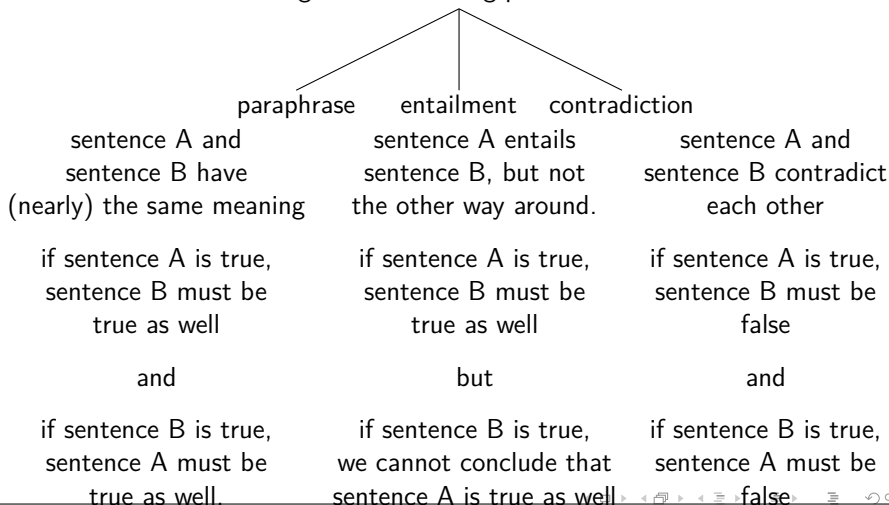
(60) It's getting hot in here.

(61) It's getting cold in here.



Meaning relations among sentences

Meaning relations among pairs of sentences





Sentence Meaning and Syntax

- ▶ The meaning of a sentence depends on how words can be combined. Accordingly, syntax and lexical semantics are essential for the correct interpretation of sentence meaning.
- ▶ Although sentences may consist out of the same words, their meaning can still vary.
- ▶ Example (subject and object are exchanged)
 - (62) Eric kills Kenny.
 - (63) Kenny kills Eric.
- ▶ In such cases the "Principle of Compositionality" proves to be valid.



Limits of Compositionality

- ▶ The Principle of Compositionality has its limits
 - (64) Kenny has passed away.
 - (65) Kenny bites the dust.
- ▶ Both sentences have the same meaning: Kenny is dead.
- ▶ However, the meaning cannot immediately be derived from the individual meanings of the words.
- ▶ Fixed phrases such as to pass away or to bite the dust are called idioms (or idiomatic phrases).
- ▶ Like exocentric compounds, the meaning of idioms cannot be predicted from the words (morphemes) they are made up of.



Presuppositions

- ▶ Understanding the meaning of sentences depends not only on the semantic and syntactic properties of sentences, but also on the assumptions and beliefs of hearer and listener.
- ▶ Such assumptions and beliefs are called presuppositions; e.g. the sentence "The major of Boston is in town" presupposes the sentence "There is a major of Boston".
- ▶ Presuppositions differ from entailments in that presuppositions also hold true when the presupposing sentence is negated.



Presuppositions

- ▶ Words that trigger certain assumptions are called presupposition-triggers.
- ▶ The expression <managed> in the following example:

(66) She managed to open the door.

presupposes the sentence

(67) She tried to open the door.

- ▶ Since presuppositions play an important role in the correct interpretation of meaning in context, they are on the boundary between semantics and pragmatics.



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Semantics: Prototypes and Categorization

Categorization

- ▶ Understand something as being part of a group.
- ▶ Categories reflect our interpretation of the world.
- ▶ Categorization $\neq$ Classical feature based semantics

man	woman	boy	girl
[+ <i>human</i>]	[+ <i>human</i>]	[+ <i>human</i>]	[+ <i>human</i>]
[+ <i>adult</i>]	[+ <i>adult</i>]	[− <i>adult</i>]	[− <i>adult</i>]
[+ <i>male</i>]	[− <i>male</i>]	[+ <i>male</i>]	[− <i>male</i>]



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ Let's do a little experiment!
 - ▶ Take out a sheet of paper and a biro (ball pen).
 - ▶ On the following slides, you will see one term denoting a category per slide.
 - ▶ As soon as you see the term, come up with 5 members belonging to that category and write them down in German (write those members down that come to (your) mind first).



Semantics: Prototypes and Categorization

VOGEL



Semantics: Prototypes and Categorization

WERKZEUG



Semantics: Prototypes and Categorization

MÖBELSTÜCK



Semantics: Prototypes and Categorization

FARBE



Semantics: Prototypes and Categorization

GEFÜHL



Semantics: Prototypes and Categorization

(in English)

PREPOSITION



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ Let's do a little experiment!
 - ▶ Take out a sheet of paper and a biro (ball pen).
 - ▶ On the following slides, you will see one term denoting a category per slide.
 - ▶ As soon as you see the term, come up with 2 members belonging to that category and draw them (draw those members down that come to (your) mind first).



Semantics: Prototypes and Categorization

DREIECK



Semantics: Prototypes and Categorization

RECHTECK



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ All of you had ...
 - ▶ *Amsel* or *Spatz* for *Vogel* OR
 - ▶ *Hammer* or *Schraubenzieher* for *WERKZEUG* OR
 - ▶ *Tisch* or *Stuhl* for *MÖBELSTÜCK* OR
 - ▶ *Rot* or *Blau* for *FARBE* OR
 - ▶ *Liebe* or *Angst* for *GEFÜHL* OR
 - ▶ *to* or *of* for *PREPOSITION*...



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ Those of you who did not have any of these terms are weird and will probably become serial killers (jk)!



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ With respect to DREIECK and VIERECK ...
 - ▶ I predict that most of you have drawn shapes that do not have an angle less than 15 degrees OR one side being more than 3 times the length of another.



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ Since about the 1980s language is understood as part of our cognitive ability through which we organize and classify all aspects of our experience.
- ▶ Cognitive semantics is based upon the assumption that meaning is linked to the way we mentally group all kinds of perceptions and phenomena into conceptual categories.
- ▶ Categorization and conceptualization are based on the comparison of new things with ones we already know and the resulting cognitive construction of similarities between different entities.



Semantics: Prototypes and Categorization

Sprachspiele (L. Wittgenstein)

- ▶ Definition of GAME!
 - ▶ opponents
 - ▶ winners and losers
 - ▶ played for fun
 - ▶ need luck or skills
- ▶ There are "better" and "worse" games (depending on the number of criteria/features they possess)



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ Let's do a little experiment!
 - ▶ Please sit in pairs of two and take out a smart phone. One of you will take the time and note down some reaction times and another person will indicate - by raising his or her hand - whether a string of letter appearing on the screen is a real word or a nonce word (made up word).
 - ▶ On the following slides, you will see a picture and then a string of letters. Please raise your arm right arm, if it is a real word, and raise your left arm, if it is a nonce word.
 - ▶ The other person will take the time and note it down between the string of letters appearing to the arm being raised.



Semantics: Prototypes and Categorization



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Semantics: Prototypes and Categorization

Wetwam



Semantics: Prototypes and Categorization





Semantics: Prototypes and Categorization

Honig



Semantics: Prototypes and Categorization





Semantics: Prototypes and Categorization

Würfel



Semantics: Prototypes and Categorization





Semantics: Prototypes and Categorization

Tallisch



Semantics: Prototypes and Categorization





Semantics: Prototypes and Categorization

Rauter



Semantics: Prototypes and Categorization





Semantics: Prototypes and Categorization

Blätter



Semantics: Prototypes and Categorization





Semantics: Prototypes and Categorization

Bautrat



Semantics: Prototypes and Categorization

Prototypicality

- ▶ High degree of prototypicality
 - ▶ goodness-of-fit (features)
 - ▶ speed of verification
 - ▶ priming



Semantics: Prototypes and Categorization

Cognitive Semantics (Prototype Semantics)

- ▶ Some concepts of expressions like the PRESIDENT OF THE UNITED STATES are rather clear-cut, while other concepts like those of STRONG or TALL are rather fuzzy.
- ▶ Fuzzy concepts are characteristic of the human conceptual system.
- ▶ Many concepts do not only have fuzzy boundaries, but can also be graded according to their typicality.
- ▶ For example, *robins* and *sparrows* are more birdlike, i.e. more prototypical than *penguins* or *ostriches*.
- ▶ Prototypes, such as *robins* for the concept BIRD are cognitive reference points.

Semantics: Prototypes and Categorization

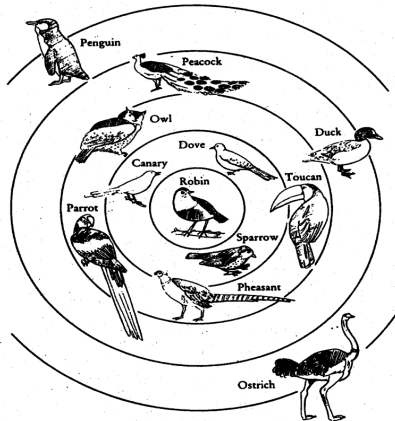


Figure 5.1 Birdiness rankings



Semantics: Prototypes and Categorization

Semantic Field Theory



Semantics: Prototypes and Categorization

Semantic Field Theory

- ▶ Let's do a little experiment!
 - ▶ I need a volunteer!



Semantics: Prototypes and Categorization

Semantic Field Theory

- ▶ Semantic field theory is based on the structuralist approach towards semantic analysis
- ▶ Structuralist approaches are based upon the assumption that every linguistic element is defined by a network of meaning relations.
- ▶ In the 1930s the structuralist assumption was applied to a new approach called Semantic Field Theory (SFT).



Semantics: Prototypes and Categorization

Semantic Field Theory

- ▶ The basic assumption of SFT is that words don't exist in isolation, but form so-called semantic fields /lexical fields together with other related words.

"The meaning of one term is defined by its relation to other terms."

- ▶ For example, color terms like "red" and "blue" are related, because they both belong to the semantic field "color".



Semantics: Prototypes and Categorization

Frame Semantics



Semantics: Prototypes and Categorization

Frame Semantics

- ▶ The basic assumption of Frame Semantics is that words and utterances are linked to behavior.
- ▶ Behavior is structured by so-called frames (or scripts)
- ▶ For example, the word "purchase" is part of the frame "financial transaction" which also involves someone who buys something, someone who is selling something, some transfer of money, and some transfer of goods.



Semantics: Prototypes and Categorization

Frame Semantics

- Words trigger frames and are understood with respect to these frames.

(68) We approached the shore.

(69) We approached the coast.

(70) Captain Miller stayed on the ground.

(71) Captain Miller stayed on the land.



Semantics: Prototypes and Categorization

Frame Semantics

- ▶ Frames are culture specific and reflect cultural practices
- ▶ Collocates of "washing" in Tanzanian and British English.
- ▶ Basic sentence types reflect Semantic Frames (think of the ditransitive construction which is related to the Semantic Frame of a TRANSACTION).



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ What is a metaphor? (provide an example)
- ▶ In which texts do you typically find metaphors?



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ Another important notion in cognitive semantics is the assumption that concepts are interconnected.
- ▶ In many cases, one concept can be understood in terms of another. This type of interconnection is called "Metaphor".

The essence of metaphor is understanding one thing in terms of another. (Lakoff and Johnson 1980)



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ In Linguistics, metaphors are a part of the conceptual system that is shared by all human beings.
- ▶ Metaphors are so very common that their use isn't even noticed
- ▶ In everyday language time is frequently treated as a valuable commodity or as a spatial property.

(72) Invest your time profitably and study linguistics!

(73) Back in the days!

(74) The day before yesterday ...

(75) Who won the argument?



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ Example: talking about arguments . . .

(76) She attacked every weak point in my argument.

(77) He couldn't defend his points.

(78) They got a lot of flak for their proposal.

(79) That argument is weak.

- ▶ Being in an argument is viewed (and talked about/conceptualized) as being in a WAR.
- ▶ this is the "WAR for ARGUMENT" or "ARGUMENT as WAR" metaphor



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ One term is taken from a source domain (typically the concrete, physical sphere) and applied to a target domain (typically a more abstract or psychological sphere), but only certain features of the concrete meaning are applied to the abstract (metaphorical) meaning.
- ▶ Metaphorical mapping

SOURCE DOMAIN	mapping	TARGET DOMAIN
Fighting parties	→	Participants
Attacking	→	Raising objections
Defending	→	Maintaining one's opinion
Surrendering	→	Giving up one's opinion



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- Task: identify the conceptual metaphors underlying the sentences below and explain why.

(80) The Iraqi democracy is still in its infancy.

(81) Only a stable, prolonged US troops presence and a weak Iraqi army will allow us to nurture democracy.

(82) The correct US policy is thus to stay the course and provide security until Iraqi democracy is strong enough to survive on its own.



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ Example: SPACE for TIME metaphor

(83) I'll meet you at five o'clock.

(84) What lies ahead ...

(85) Looking back, I regret nothing!

(86) I'll work until five o'clock.

SOURCE DOMAIN	mapping	TARGET DOMAIN
locations	→	specific times
path not yet trodden	→	future
path already walked	→	past
spatial barriers (walls)	→	end/beginning of timespan



Semantics: Prototypes and Categorization

Cognitive Semantics (Metaphor)

- ▶ Metaphors enable communication about abstract concepts by allowing us to talk about one thing in terms of another
- ▶ Metaphors work by mapping cognitive structures from a source domain to a target domain.
- ▶ Metaphors help us understand/make sense of the world around us.



Semantics: Prototypes and Categorization

Metonymy

- ▶ Metonymy concern mapping within domains not between domains (metaphor)

(87) How much is the Picasso?

(88) I like to read Camus.

(89) Shakespeare is on the top shelf.

- ▶ PRODUCER → PRODUCT

(90) I see a lot of new faces.

(91) We need a couple of strong bodies for that.

(92) Google hires the best brains.

- ▶ PART OF PERSON → WHOLE PERSON



Semantics: Prototypes and Categorization

Metonymy

- ▶ Metonymy concern mapping within domains not between domains (metaphor)

(93) The guitar is too loud.

(94) She was in tears.

(95) The movie was sad.

- ▶ CAUSE → EFFECT

(96) Pearl Harbor still affects current US policy.

- ▶ PLACE → EVENTS

(97) Germany beat Brazil in the World Cup.

- ▶ COUNTRY → REPRESENTATIVES OF A COUNTRY



Semantics: Prototypes and Categorization

Metonymy

- ▶ Metonymy is referring to one entity by means of another related entity.
- ▶ Two entities belong to the same semantic domain.
Semantic shift within the same semantic category.
Metonymy follows principles that can be stated as ...
PART FOR WHOLE
CAUSE FOR EFFECT
PRODUCER FOR PRODUCT
...



Semantics: Prototypes and Categorization

Metonymy

- Identify the metonymies in the following sentences.

(98) France opposed the war in Iraq.

(99) Spaghetti are my favorite dish.

(100) The ham sandwich left without paying.

(101) This weekend, Britain goes to the ballot.



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Pragmatics

Today's topics and terms

- ▶ Extra-sentential meaning
- ▶ Deixis
- ▶ Context
- ▶ Coreferentiality
- ▶ Speech acts
- ▶ Conversational implicatures



Pragmatics

What is pragmatics?

- ▶ Pragmatics examines how speakers understand and communicate more than the literal meaning of words and sentences. The type of meaning studied in pragmatics is known as utterance meaning, meaning in context or meaning in interaction. In short, pragmatics is about getting from what is said to what is meant.

(Becker and Bieswanger 2006: 161)



Pragmatics

What is pragmatics?

- ▶ Focuses on linguistic performance (parole) rather than on the underlying linguistic system (langue).
- ▶ In a narrow definition, pragmatics is regarded as a sub-discipline of semantics, i.e. it deals with meaning in communication. Communication is conceived of as negotiation of meaning between interlocutors.
- ▶ Meaning is mostly implicit. It is the goal of the hearer to recognize the speaker's communicative intention.



Pragmatics

Pragmatic turn

- ▶ Pragmatics is commonly contrasted with structuralist or formal approaches
- ▶ Formal approaches focus on language systems (langue) and disregard language external factors.
- ▶ During the 1970s, linguistics became very formal (Chomsky's generative grammar; context independent; syntactic rules)
- ▶ Formal approaches were criticized for their neglect of the speaker
- ▶ Consequence, some linguists turned to the context and functions that language was used in and for.



Pragmatics

What is pragmatics?

- ▶ Pragmatics is the study of meaning in context.
- ▶ The same sentence can have different meaning depending on the context in which it is uttered.

(102) A police man is right around the corner!
(Two criminals discuss whether or not to rob a store)

(103) A police man is right around the corner!
(A tourists asks someone for directions)

(104) A police man is right around the corner!
(A person being threatened by someone)



Pragmatics

Code model of communication

Speaker : thought $\xrightarrow{\text{encoding}}$ message $\xrightarrow{\text{decoding}}$ thought : Hearer



Pragmatics

Problems of the *code model of communication*

- ▶ Disregards context.
- ▶ Neglects the role of inference.



Pragmatics

What is context?

- ▶ Context makes things "go without saying"
- ▶ Context is always available (as a resource)
- ▶ Meaning is (almost) always context-bound

Aspects of context

- ▶ Physical environment (e.g. place deixis)
- ▶ Participants (e.g. person deixis)
- ▶ Social setting (e.g. who you are talking to and where you are talking to them)
- ▶ Prior discourse (discourse deixis)
- ▶ Cultural norms/expectations (e.g. politeness; face-saving/directness)



Pragmatics

What is Inference?

- ▶ Conveyed message goes beyond/is not identical to what is literally said
- ▶ Hearers infer what is meant (what is intended).
- ▶ The message is enriched by the hearer.
- ▶ Relying on inference is the norm in language.



Pragmatics

What is Inference?

- (105) Since Bob runs the coffee shop, the espresso tastes a bit funny.
- (106) Bob is poor but honest
(careful! relies on social stereotypes: poor people are typically not honest)
- (107) The king had a heart attack and a republic was declared.
- (108) A republic was declared and the king had a heart attack.



Pragmatics

What is said What is meant
What does X mean? → What do you mean by X?

- ▶ Locution: what is said
- ▶ Illocution: what is meant/intended



Pragmatics

Direct Speech Acts

(109) Please clean the dishes!

(110) I bet you 5 € that the HSV will win.

(111) I hereby declare you husband and wife.

- Locution and illocution coincide.



Pragmatics

Utterance

- ▶ Pragmatics has been defined as scientific study of the interpretation and production of utterances.
- ▶ Utterances are spoken, written or gestured contributions whose meaning depends on the social context of the speech situation.
- ▶ In spoken language analysis an utterance is a smallest independent unit of speech.



Pragmatics

(112) The telephone is ringing. . .

A: The telephone is ringing!

B: I'm in the bath!

A: Okay, okay . . . I'll get it . . .

- ▶ Speaker A does not simply say that the telephone is ringing and speaker B does not merely say that he/she is in the bath.
- ▶ Speaker A's utterance is a request and speaker B's reply provides a reason why he/she cannot comply with this request.



Pragmatics

Micropragmatics (conversational pragmatics)

- ▶ Micropragmatics allows us to bridge the gap between the propositional meaning and the meaning a utterance has within a certain context (the gap between what is said and what is meant).
- ▶ Pragmatics = meaning minus semantics.



Semantics and Pragmatics

Semantics	Pragmatics
context invariant	context sensitive
speaker independent meaning	speaker dependent meaning
meaning potential	concrete meaning
What does X mean	What does speaker A mean by uttering X
conventional meaning	non-conventional meaning
what is said	what is meant
Description of meaning, meaning relations, and meaning combinations	Bridging the gap between what is said and what is meant



Intercultural Pragmatics

- ▶ Language use and understanding of utterances is to a certain extent language specific
- ▶ Linguistic communities differ according to what is said how when and why to whom about someone else.
- ▶ Knowing the semantic and syntactic properties of a language is not sufficient for real life communication.
- ▶ What is missing?
 - ▶ Pragmatic competence
 - ▶ Ability to use language appropriately within social contexts



Deixis and Context

- ▶ Imagine you find a piece of paper with the note:
(113) I'll see you there tomorrow.
- ▶ What do <I>, <you>, <there>, and <tomorrow> refer to?
- ▶ Greek *deiknynai*: "to point"/ "to show"
- ▶ Pointing to extra-linguistic contexts
- ▶ Deictic expressions
 - ▶ Also called deictics or indexicals (lat.: *indicare*)
 - ▶ Expressions that are used to point at someone or something
 - ▶ Meaning is context dependent



Deixis and Context

- ▶ Person deixis

- ▶ Referring to persons (I, you, she, he, it, etc.)

Place deixis

- ▶ Referring to locations (here, there, left, right, ect.)

Time deixis

- ▶ Referring to points or stretches of time (today, tomorrow, etc.)

Social deixis

- ▶ Reflecting social structures (dt.: Sie vs. Du)

Discourse deixis

- ▶ Refers to activities within texts (... as mentioned in section 3)



Deixis



The ball is to the right of the car. The ball is behind the car.



Deixis and Context

- ▶ Deictic Center (Origo; Karl Bühler 1934)
 - ▶ When we speak to other people we tend to take their perspective
 - ▶ Perspective dependency
 - ▶ Ego of the other person become the deictic center. This is called deictic projection
 - ▶ Deictic projection
 - ▶ Identifying the deictic center is crucial for the correct interpretation of all deictic expressions
 - ▶ Important for place deixis, especially



Social deixis

T/V distinction

- ▶ Lat.: tu (2nd pers. singular) vs. vos (2nd pers. singular and plural)
- ▶ Dt.: Du (2nd pers. singular) vs. Sie (2nd pers. singular and plural)
- ▶ Important for the right choice of form of address
- ▶ First name, surname, title, office?
 - ▶ John?
 - ▶ Mr. Smith?
 - ▶ Dr. Smith?
 - ▶ Sergeant Smith?



Place deixis

"Pointing" at a location

(114) You *there* in the last row!

- ▶ Closeness and Distance

- ▶ English and German distinguish between entities close to the origo and entities that are further away.
- ▶ Proximal terms
 - ▶ here/ hier and this/ dieses
- ▶ Distal terms
 - ▶ there/ da, dort and that/ jenes
 - ▶ often used to express psychological distance

(115) ... *that* boring book ...

(116) ... *that* awful lesson ...



Time deixis

"Pointing" at a point in time or a time span.

(117) Martin was here *yesterday*!

- Terms used to refer to time are commonly adopted from the terminology used to refer to places and spatial relations.

(118) now, soon, then, ago, today, yesterday, tomorrow, present, actual, current, former, future, next, last.



Coreferentiality

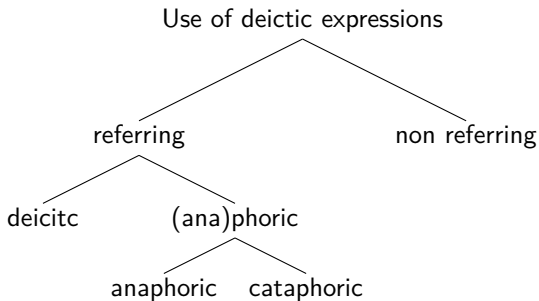
- ▶ Some deictic expressions, i.e. anaphoric expressions are used to refer not deictically, but refer to entities that have been introduced before or will be introduced later on.
- ▶ Anaphora
 - ▶ Referents of a anaphoric expressions have been introduced before.

(119) Sally went to the pub, and she only went home when the sun had started to rise.

- ▶ Cataphora
 - ▶ Referent of a cataphoric expressions are introduced later.
- (120) He saw her, but only seconds later George recognized her.



Coreferentiality





Philosophy of meaning

- ▶ Pragmatics was not introduced by linguists but by proponents of "philosophy of language".
- ▶ Certain philosophers were dissatisfied by the application of formal logic to the analysis of natural languages.
 - ▶ In truth- conditional semantics, the fact that meaning was context-dependent was ignored.
 - ▶ Statements were analyzed as propositions that had truth conditions (under which conditions is a statement true or false) and truth values (being either true or false).
 - ▶ This critique was put forward by the late Wittgenstein, Searle, Austin, and Grice who regarded themselves as a counter movement to traditional logic which ignored central aspects of natural language and ordinary communication.



Speech Acts

- ▶ The "new" school of philosophers of language introduced two important pragmatic theories: "Speech Act Theory" and the "Theory of Conversational Implicatures".
- ▶ Speech Act Theory (Austin and Searle)
 - ▶ Language does not (solely) consist of statements about the world and cannot be assessed in terms of truth conditions.
 - ▶ Truth conditions are not useful in order to understand the underlying intentions of the speaker.

(121) Happy birthday!

(122) Merry Christmas!

(123) Will you go to the dance with me?

(124) I hereby declare the meeting closed!



Speech Acts

- ▶ The relevant information, i.e. the actual message is implicit, not explicit!
- ▶ The relevant aspects of meaning which are relevant to truth-conditional semantics under-specify the message conveyed.
 1. Communication is a dynamic process.
 2. To communicate is to "act".
 3. Communication is successful, only if the hearer grasps the intended meaning.
- ▶ Verbal interaction is defined as a "speech act".
- ▶ Speech acts can differ in respect to their illocution (=intended meanings), i.e. the same message can be intended to be a threat, a compliment, a request, ...



Types of Speech Acts

- ▶ Assertives (representatives)
 - ▶ Used to describe the world (state, claim, describe, ...)
- ▶ Directives
 - ▶ Used to get people to do things
(give an order, or ask so. to do. sth., ...)
- ▶ Commissives
 - ▶ Express commitment to a future action
(promising, threatening, ...)
- ▶ Expressives
 - ▶ Employed to express feelings and opinions
(thank, greet, apologize, ...)
- ▶ Declarations
 - ▶ Used to express a change of affairs
(baptisms, marriages, divorces, declarations of war, ...)



Types of Speech Acts

- ▶ Direct and indirect speech acts
 - ▶ All speech acts are either "direct", i.e. explicit (secondary) or "indirect", i.e. implicit (primary).
 - ▶ Implicit requests are performed by means of direct speech:

(125) It is freezing in here!

- ▶ The implication being the indirect request "Please shut the window!".
- ▶ The primary intention is commonly the implied one (hence, the primary/secondary distinction).



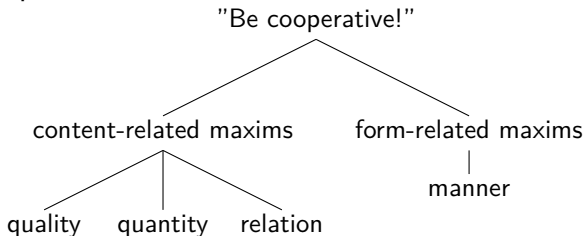
Conversational Implicatures

- ▶ Grice invented the notion of "conversational implicatures".
- ▶ Pragmatic inferences request not only information of semantic meaning but also require some knowledge about the context.
- ▶ The same utterances can have quite different pragmatic inferences if uttered in different contexts (semantic inferences are context-independent).
- ▶ "Cooperative Principle"
 - ▶ Make your conversational contribution such as required, at the stage at which it occurs, by the accepted purpose or direction of the talk exchange in which you are engaged!



Conversational Implicatures

- Communication is cooperative behavior, it underlies the assumption that speaker and hearer want to co-operate and primarily consists in asking oneself: what does the speaker mean?





Conversational Implicatures

- ▶ Quality
 - ▶ Make your contribution one that is true. Do not say what you believe to be false (Quality₁) and not more informative as required (Quality₂).
- ▶ Quantity
 - ▶ Make your contribution as informative as is required for the current purposes of the exchange (Quantity₁).
- ▶ Relation
 - ▶ Be relevant. Do not change the topic.
- ▶ Manner
 - ▶ Be prespicious: avoid obscurity of expression (Manner₁), avoid ambiguity (Manner₂), be brief (Manner₃), and orderly (Manner₄).



Conversational Implicatures

- ▶ The maxims of conversational implicatures are descriptive and reflect everyday behavior.
- ▶ The maxims of conversational implicatures are not the only principles relevant for communication. Other principles that are highly relevant are principles like principles of politeness or of face-saving.
- ▶ One important aspect of implicational implicatures is cancellability, i.e. they can be canceled without a sense of contradiction.

(126) I need someone who speaks Russian OR Polish!

(127) Of course anyone speaking both languages is welcome.



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Sociolinguistics and Language Change

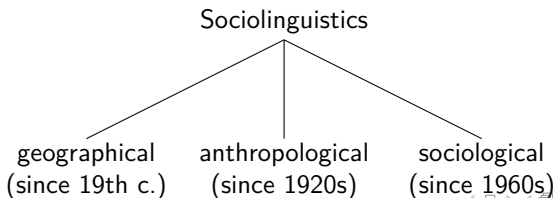
Today's topics and terms

- ▶ Varieties
- ▶ Ordered heterogeneity
- ▶ Apparent time and real time
- ▶ Dialectology



Sociolinguistics

- ▶ Sociolinguistics
 - ▶ "Its aim is to study the use of different forms or varieties of language and the social factors which determine them." (Kortmann 2001: 254)
- ▶ Sociology of language
 - ▶ "Its motivation for investigating language is to increase the ability to understand social structures." (Kortmann 2001: 254)



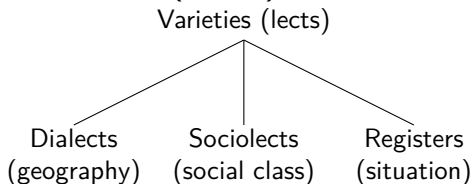


Types of sociolinguistics

- ▶ Geographical sociolinguistics/Dialectology (since 19th c.)
 - ▶ studies the variation in the phonological and lexical system of NORMs (non-mobile, old, rural males) using questionnaires, word lists and tape recording; isoglosses (boundaries on language maps)
- ▶ Anthropological sociolinguistics (since 1920s)
 - ▶ studies the relationship between language, culture and thought; e.g. to which extent do linguistic structures determine thought and ultimately perception (Sapir-Whorf Hypothesis)
- ▶ Sociological sociolinguistics (since 1960s)
 - ▶ studies the variation of language use with regard to sociological variables such as gender, age, ethnicity, etc., by employing quantitative methodology (statistics)



Types of varieties (lects)

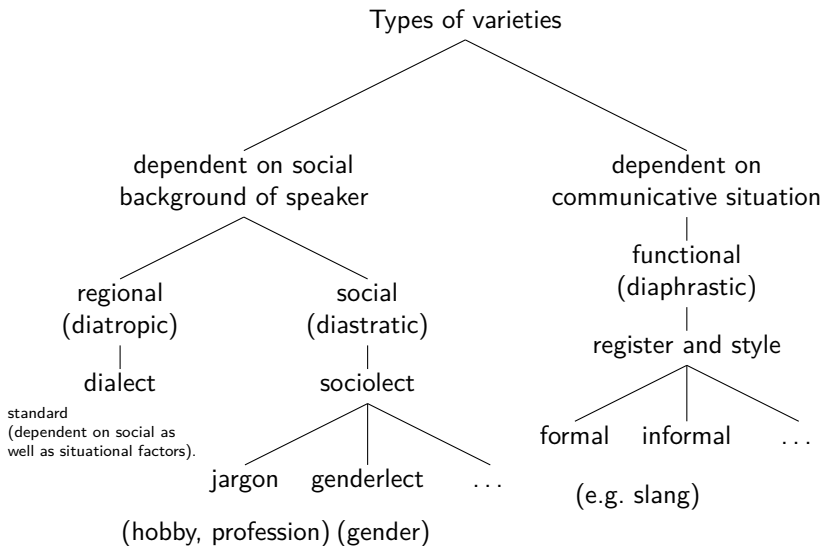


- ▶ Dialects
 - ▶ regionally restricted varieties (narrow sense)
 - ▶ same as 'variety': social dialect (broad sense)
- ▶ Sociolects
 - ▶ motivated by socio-economic status, level of education, profession, age, ethnicity, sex, ... (genderlect, jargon)
- ▶ Register (only vocabulary; if also grammar: style)
 - ▶ varieties determined by communicative situation



Types of varieties (lects)

- ▶ Varieties versus "the Standard"
 - ▶ From a linguistic point of view, they are different, but not inferior, because the structural properties of any variety have the same value or quality as any other variety.
 - ▶ Standard varieties, however, do enjoy higher prestige.
 - ▶ Standard varieties are used in education, broadcast, written language, politics, courts, ...
 - ▶ Varieties become "standard" because of social and functional considerations NOT because of structural (linguistic) properties!
 - ▶ Standard ~ common structural core of various different lects.





Sociolinguistics

- ▶ Language communities are not homogeneous; instead heterogeneity determines everyday language use.
- ▶ Not only do different groups within language communities differ, but even each speaker has his or her own idiolect.
- ▶ According to the communicative situation and the speaker's expressive ability, all language users chose different language forms – the different form of a language are called varieties (or lects).



Sociolinguistics

- ▶ One speaker social belongs to different groups, thus has many selves, i.e. exists within a diversity of co-existing group identities and uses language according to the role one fulfills.
- ▶ Sociolinguistics is the study of which factors (variables) determine which linguistic form and describe which features a variety possesses or lacks.



Sociolinguistics

Varieties

- ▶ Group identities are formed by geographical, social and/or ethnic background, but other factors such as age, sex, profession and level of education also influence linguistic performance.
- ▶ Few words suffice to make us form inferences about a speaker's social background, level of education, etc and signal that he or she "wants" to belong/belong to a certain social group.



Sociolinguistics

Varieties

- ▶ How do the terms "variety" and "dialect" relate to "standard" and "language"?
 - ▶ Dialects and varieties are not inferior but they are of no higher value or quality than other forms of a language.
 - ▶ Nevertheless, some (standard) varieties enjoy higher prestige.
 - ▶ The characterization of one variety as the standard variety of a language is based on social and functional considerations, not on linguistic considerations.



Sociolinguistics

Varieties

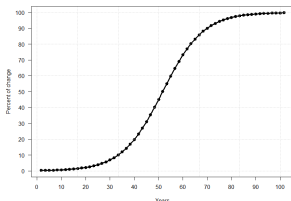
- ▶ Standards represent the "common structural core" of all varieties of a language. This common core is relatively homogeneous, i.e. there are few regional differences in the standard varieties of geographical varieties (Old Englishes vs. New Englishes).
- ▶ Due to its special functional role, the standard may be regarded as a fourth (main) type of variety.
- ▶ In all language communities, the standard variety is only spoken by a small number of speakers.
- ▶ The standard variety commonly serves as a reference variety if varieties are compared.



Variation and change

Varieties

- ▶ How do we bridge the gap between synchrony and diachrony?
 - ▶ Variation among certain social groups is adopted by other groups.
 - ▶ Innovations spread from an epicenter outward and diffuse through the lexicon and the speech community.

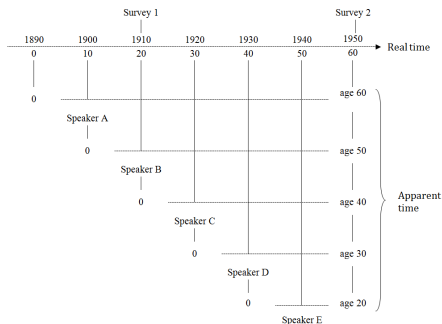




Variation and change

Apparent time and real time

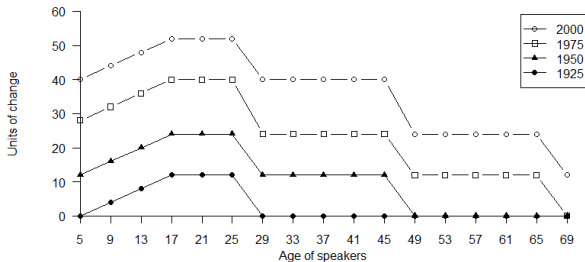
- Changes can be detected and analyzed by looking at language use by different speaker groups at different times/of different ages.





Variation and change

Apparent time and real time





Variation and change

Patterns of change

	Individual	Community
Stability	Stable	Stable
Age-grading	Unstable	Stable
Generational change	Stable	Unstable
Communal change	Unstable	Unstable



Variation and change

Gender and change

- ▶ Most of the linguistic changes in progress studied in the 2nd half of the 20th century show a high degree of social differentiation. (Labov 2002)
- ▶ Gender Paradox
 - ▶ Women conform more closely than men to sociolinguistic norms that are overtly prescribed, but conform less than men when they are not. (Labov 2001: 293)
- ▶ Male-dominated changes form a "small minority". (Labov 2001: 284)



Variation and change

Gender and change

- ▶ Male-dominated changes form a "small minority". (Labov 2001: 284)

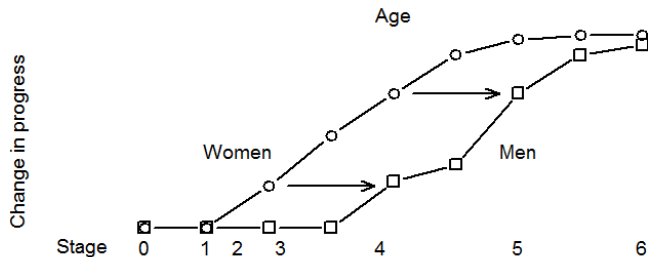




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Recapitulation

- What have we done?!?



Berg, T. (2013). *Anglistische Sprachwissenschaft*. Munich: Wilhelm Fink GmbH and Co. Verlags-KG.

Kortmann, B. (2001). *Linguistik: Essentials, Anglistik, Amerikanistik*. Cornelsen.

Labov, W. (2001). *Principles of Linguistic Change: Social Factors*, Volume 2. Blackwell.

Labov, W. (2002). Driving forces in linguistic change. Paper presented at the International Conference on Korean Linguistics.

<http://www.ling.upenn.edu/~wlabov/Papers/DFLC.htm>.

Lakoff, G. and M. Johnson (1980). The metaphorical structure of the human conceptual system. *Cognitive Science* 4(2), 195–208.